1. Project Introduction

a) The title of this project is “Unemployment in Islington”, the project topic is based on data from real-world Islington iWork unemployment support from 2018 – 2021.

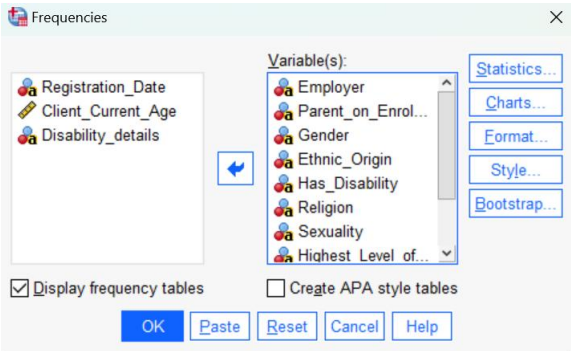
b) The aim for this coursework is to analyse the Islington iWork unemployment support data from 2018 to 2021 to identify different factors such as socio-demographic influencing employment and to offer recommendations at helping support services within the Islington borough. The objectives will include, data exploration, pattern recognition from and predictive modelling.

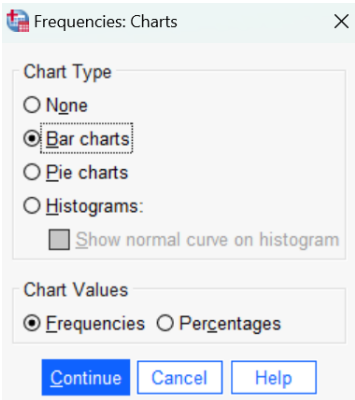
c) The methodology to be used will be as follows :
Data collection – Using the data provided by iWork and importing the data set into SPSS.
Data preparation - Cleaning / preparing the data for further analysis by identifying missing values and outliers.
Data transformation - Preparing data in numerical / categorical where it is possible.
Data analysis – Analysing the data to gain further understanding of it.

All of this will be done using SPSS.

2. Descriptive statistics analysis of each variable with :

a)





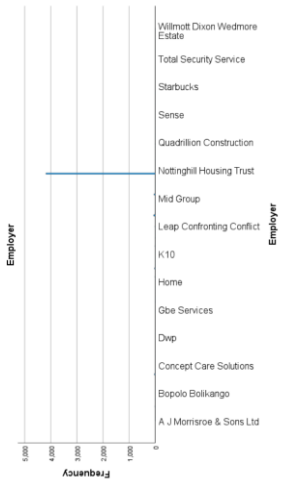
I will now display the frequency distribution with the nominal variables in a bar chart form, in my opinion a bar chart is the second best option to display the frequency distribution, first being a histogram but that cant be done on string value variables. I have left “Registration_Date”, “Client_Current_Age” and “Disability_details” out as “Registration_Date” contains dates which are best handled as a temporal variable, “Client_Current_Age” isn’t nominal and “Disability_details” contains all null values.

Frequency Distribution Charts

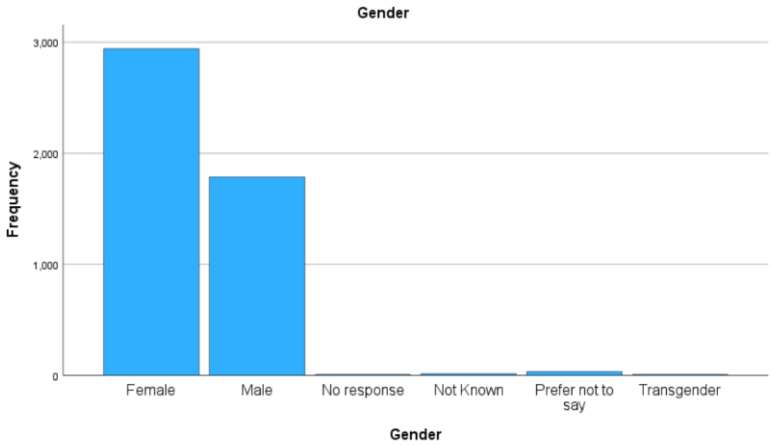
This frequency table frequency distribution for the “Employer” variable, we can see that the most frequent outcome is “No Outcome” with a frequency of 4184.

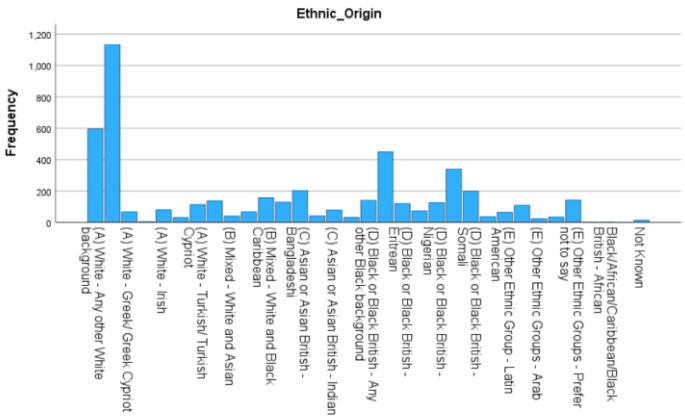
Snippet from the “Employer” frequency table

Network Homes	1	.0	.0	9.6
Next Plc	1	.0	.0	9.7
Nhs	1	.0	.0	9.7
Nhs Sutton	1	.0	.0	9.7
Nike Town	1	.0	.0	9.7
No Outcome	4184	87.4	87.4	97.1
North Islington Children's Centre	1	.0	.0	97.1
North Middlesex Hospital	1	.0	.0	97.2
Nottinghill Housing Trust	1	.0	.0	97.2
Nuffield Health	1	.0	.0	97.2
NULL	1	.0	.0	97.2
Office Of National Statistics	2	.0	.0	97.3



This bar chart shows the frequency distribution for the “Gender” variable, we can see that the most frequent gender is “Female” (2942) followed by “Male” (1787), the other gender groups make up for the remaining 59 values.



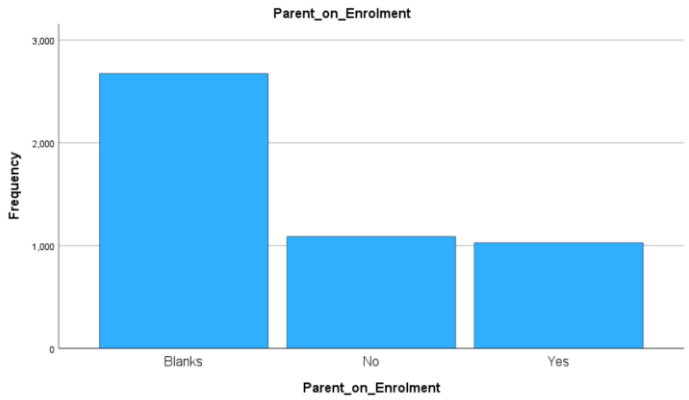


This bar chart and frequency table shows the frequency distribution for the "Ethnic_Origin" variable, we can see that the most frequent ethnic origin is "White - British" with a frequency of 1132 which makes up for 23.6% of the data.

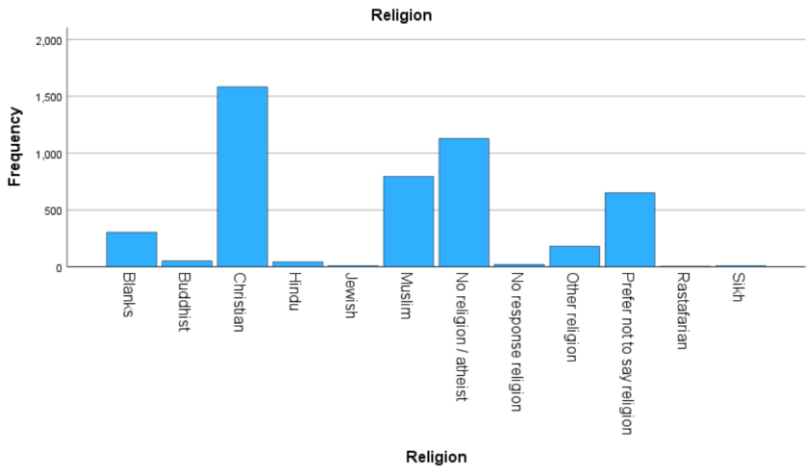
Ethnic_Origin

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	(A) White - Any other White background	595	12.4	12.4	12.4
	(A) White - British	1132	23.6	23.6	36.1
	(A) White - Greek/ Greek Cypriot	67	1.4	1.4	37.5
	(A) White - Gypsy/ Traveller	5	.1	.1	37.6
	(A) White - Irish	81	1.7	1.7	39.3
	(A) White - Kurdish	30	.6	.6	39.9
	(A) White - Turkish/ Turkish Cypriot	113	2.4	2.4	42.3

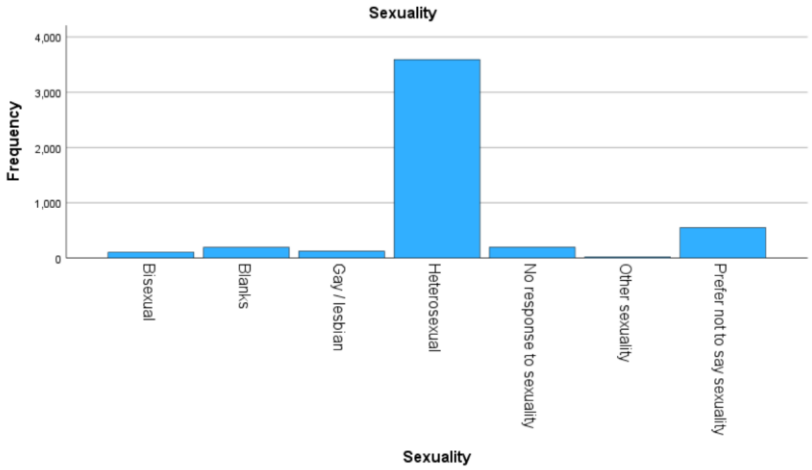
This bar chart shows the frequency distribution for the "Parent_on_Enrolment" variable, we can see that the most frequent response is "Blanks" (2673) followed by "No" (1087) then "Yes" (1028).



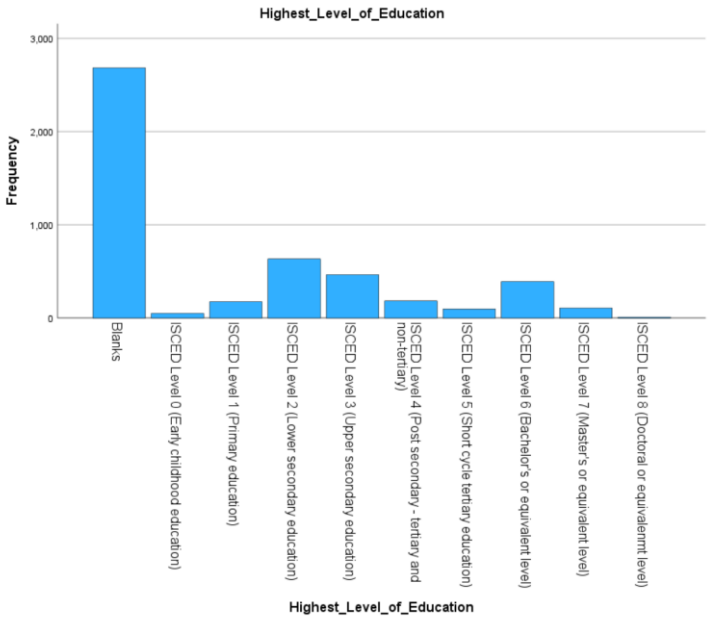
This bar chart shows the frequency distribution for the "Religion" variable, we can see that the most frequent response is "Christian" (1584) followed by "No religion / atheist" (1128).



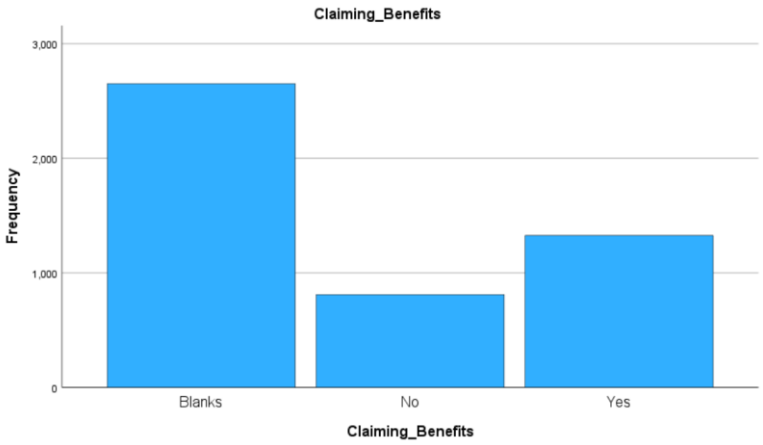
This bar chart shows the frequency distribution for the "Sexuality" variable, we can see that the most frequent response is "Heterosexual" (3592) making up 75% of the data.



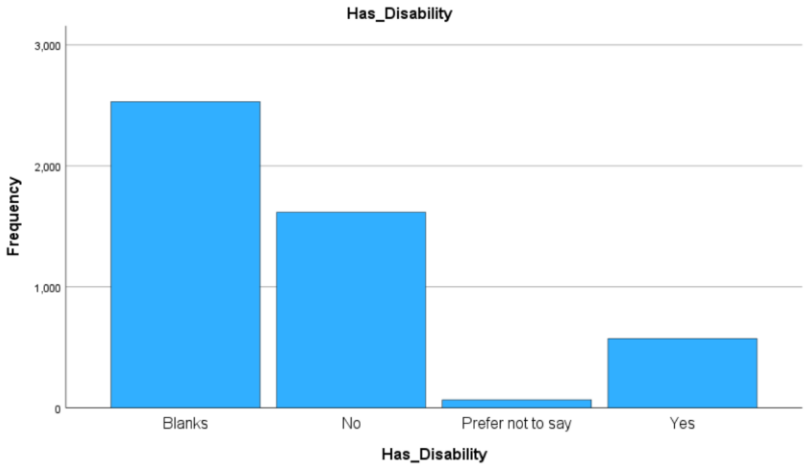
This bar chart shows the frequency distribution for the "Highest_Level_of_Education " variable, we can see that the most frequent response is "Blanks" (2684) making up for 56.1% o the data.



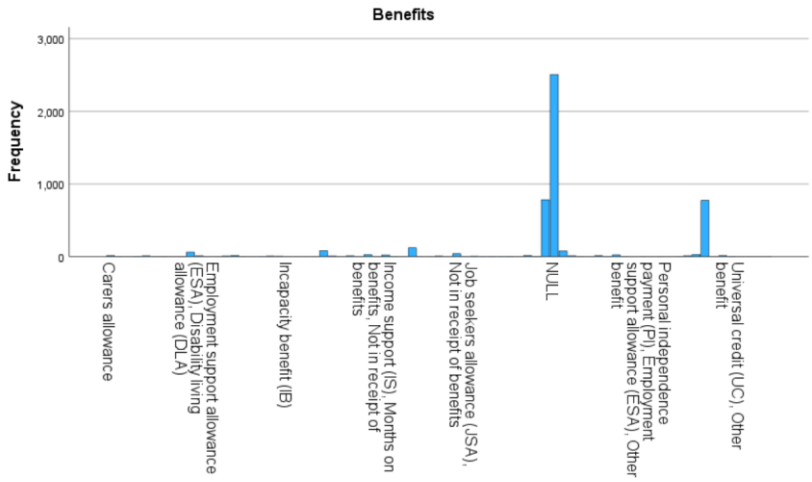
This bar chart shows the frequency distribution for the "Claiming_Benefits" variable, we can see that the most frequent response is "Blanks" (2651) followed by "Yes" (1326) and lastly "No" (811).



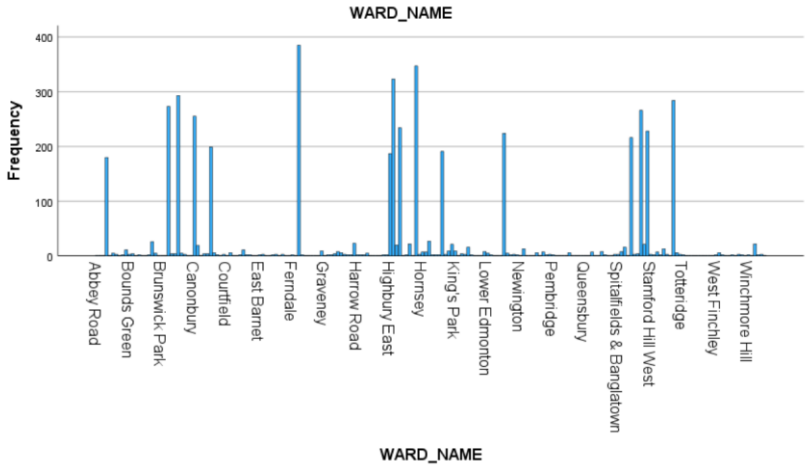
This bar chart shows the frequency distribution for the “Has_Disability” variable, we can see that the most frequent response is “Blanks” (2531) followed by “No” (1617) then “Yes” (573) lastly “Prefer not to say” (67).



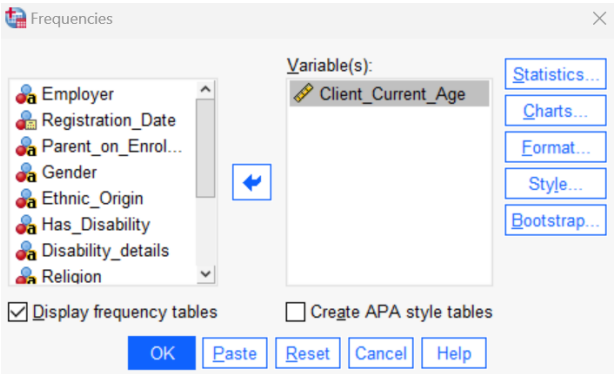
This bar chart shows the frequency distribution for the “Benefits” variable, we can see that the most frequent response is “Null” (2507) followed by “Not in receipt of benefits” (783).

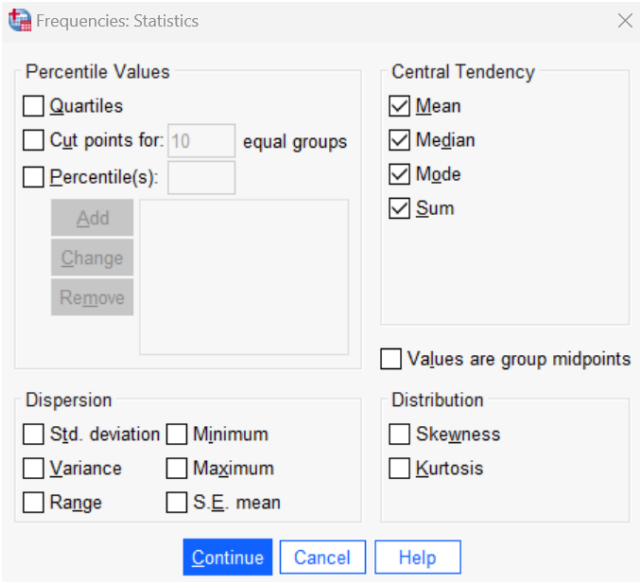


This bar chart shows the frequency distribution for the “WARD_NAME” variable, we can see that the most frequent response is “Finsbury Park” (835) closely followed by many different responses.



Central tendency and dispersion





The following image will show the central tendency and dispersion of the “Client_Current_Age” interval variable. To explain this table it shows us the amount of valid entries which are 4788, any missing values which there isn’t, the mean which is 37.33, the median 36.00, the mode 20 and the sum of the ages is 178718.

Statistics

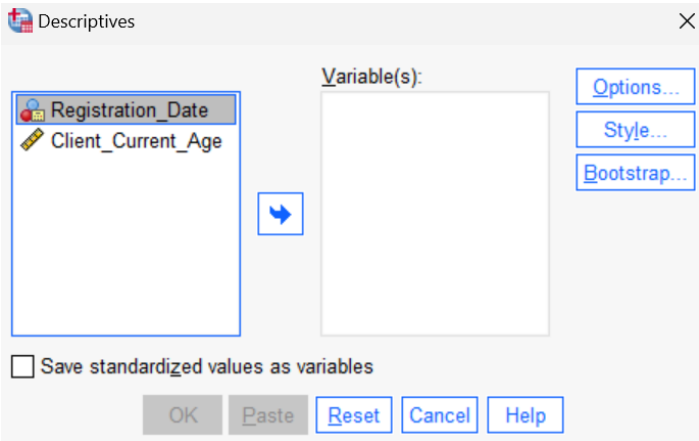
Client_Current_Age

N	Valid	4788
	Missing	0
Mean		37.33
Median		36.00
Mode		20
Sum		178718

3. Data preparation and transform for further data analysis

There are a few ways to identify the missing data, firstly we can just look at the data set and look for null values, however with bigger data sets this is not recommended.

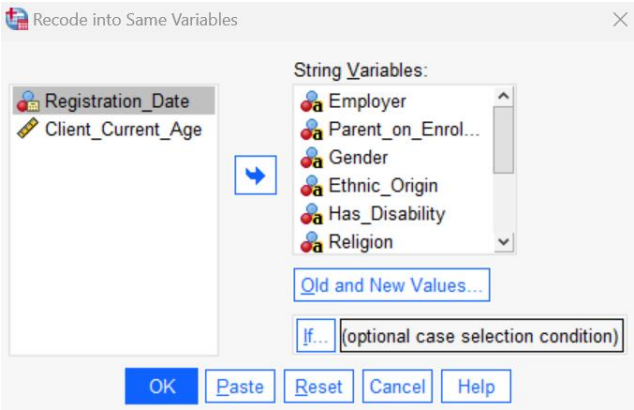
I looked through the data set to begin with and noticed a lot of null values in the “Disability_details” variable to double check my suspicions I performed the following analysis on the variable.

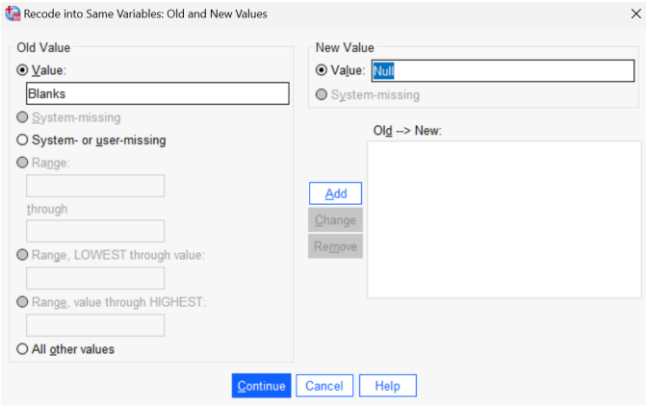


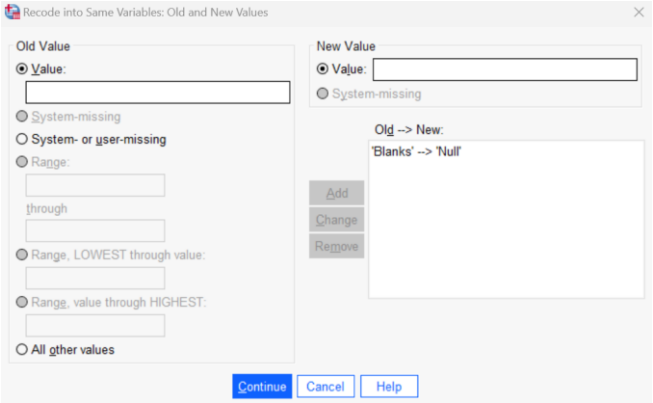
As you can see the “Disability_details” variable doesn’t show up as if SPSS reports all null values it will exclude the variable from descriptive analysis by default. I will delete the whole variable by going to “Variable view” right clicking “Disability_details” and clicking clear. The following image shows the remaining variables.

	Name
1	Employer
2	Registratio...
3	Client_Curr...
4	Parent_on_...
5	Gender
6	Ethnic_Origin
7	Has_Disabi...
8	Religion
9	Sexuality
10	Highest_Le...
11	Claiming_B...
12	Benefits
13	WARD_NA...

Now for further data cleaning I have recoded “Blanks” as NULL values, this way both “Blanks” and nulls are treated as missing values ensuring consistent analysis.





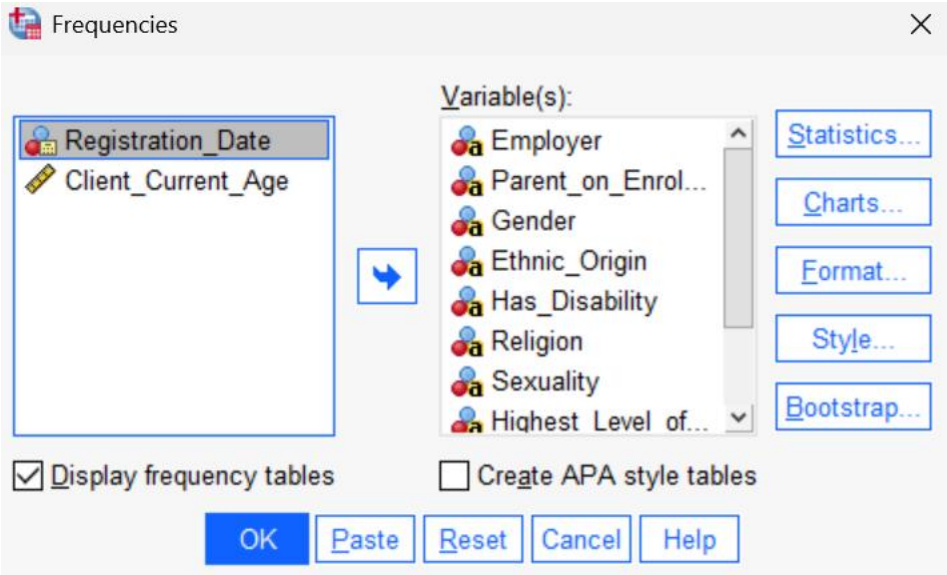


	Employer	Registration_Date	Client_Current_Age	Parent_on_E...	Gender	Ethnic_Origin	Has_Disability	Religion	Sexuality	Highest_Level_of_Education	Claiming_Be...
1	No Outcome	22-Jul-2021	29	NULL	Female	(C) Asian or Asian British - Any o...	No	NULL	Prefer not to say sexuality	ISCED Level 6 (Bachelor's or equiv...	No
2	No Outcome	24-Aug-2021	32	NULL	Male	(C) Asian or Asian British - Any o...	No	NULL	Prefer not to say sexuality	ISCED Level 6 (Bachelor's or equiv...	No
3	No Outcome	13-May-2021	48	NULL	Female	(D) Black or Black British - Other ...	NULL	Christian	Heterosexual	NULL	NULL
4	No Outcome	31-Aug-2021	55	NULL	Male	(D) Black or Black British - Any ot...	No	NULL	No response to sexuality	ISCED Level 7 (Master's or equival...	Yes
5	No Outcome	31-Aug-2021	30	NULL	Female	(A) White - Any other White back...	No	NULL	No response to sexuality	ISCED Level 2 (Lower secondary e...	Yes
6	No Outcome	31-Aug-2021	57	NULL	Male	(A) White - British	No	NULL	No response to sexuality	ISCED Level 6 (Bachelor's or equiv...	Yes
7	No Outcome	31-Aug-2021	34	NULL	Male	(C) Asian or Asian British - Indian	No	NULL	No response to sexuality	ISCED Level 7 (Master's or equival...	No
8	No Outcome	01-Jul-2021	21	NULL	Male	(C) Asian or Asian British - Any o...	No	NULL	Prefer not to say sexuality	ISCED Level 3 (Upper secondary e...	Yes
9	No Outcome	27-Aug-2021	18	NULL	Female	(C) Asian or Asian British - Bang...	Yes	NULL	Prefer not to say sexuality	ISCED Level 3 (Upper secondary e...	No
10	No Outcome	06-Apr-2021	34	NULL	Male	(D) Black or Black British - Carib...	Prefer not to say	NULL	Prefer not to say sexuality	NULL	No
11	No Outcome	15-Apr-2021	32	NULL	Female	(A) White - British	NULL	No religion / atheist	Heterosexual	NULL	NULL
12	No Outcome	18-Jun-2021	32	NULL	Female	(C) Asian or Asian British - Bang...	NULL	Muslim	Heterosexual	NULL	NULL
13	No Outcome	10-Jun-2021	33	NULL	Female	(A) White - British	NULL	No religion / atheist	Heterosexual	NULL	NULL
14	No Outcome	03-Mar-2021	29	NULL	Female	(D) Black or Black British - Any ot...	No	NULL	No response to sexuality	ISCED Level 4 (Post secondary - te...	No
15	No Outcome	10-Mar-2021	47	NULL	Female	(A) White - British	NULL	No religion / atheist	Heterosexual	NULL	NULL
16	Goodinge Group Practice	29-May-2020	40	No	Female	(A) White - British	No	Prefer not to say religion	Prefer not to say sexuality	ISCED Level 6 (Bachelor's or equiv...	No
17	No Outcome	03-Mar-2021	31	NULL	Female	(D) Black or Black British - Any ot...	No	NULL	No response to sexuality	ISCED Level 7 (Master's or equival...	No
18	No Outcome	10-May-2021	44	NULL	Female	(A) White - Turkish/ Turkish Cypri...	NULL	Muslim	Heterosexual	NULL	NULL
19	No Outcome	12-Jun-2019	36	No	Male	(A) White - Any other White back...	No	No religion / atheist	Heterosexual	ISCED Level 3 (Upper secondary e...	Yes
20	No Outcome	29-Apr-2021	43	NULL	Female	(D) Black or Black British - Carib...	NULL	Prefer not to say religion	Heterosexual	NULL	NULL
21	No Outcome	28-Nov-2019	31	Yes	Female	(A) White - British	No	No religion / atheist	Heterosexual	ISCED Level 4 (Post secondary - te...	Yes
22	No Outcome	06-Aug-2021	45	NULL	Female	(D) Black or Black British - Somali	NULL	Muslim	Heterosexual	NULL	NULL
23	No Outcome	21-Jun-2021	42	NULL	Male	(C) Asian or Asian British - Any o...	No	NULL	Prefer not to say sexuality	ISCED Level 7 (Master's or equival...	No
24	No Outcome	04-Mar-2021	47	NULL	Male	(D) Black or Black British - Carib...	NULL	Prefer not to say religion	Bisexual	NULL	NULL
25	No Outcome	21-Apr-2021	47	NULL	Male	(B) Mixed - White and Black Afric...	NULL	Christian	Bisexual	NULL	NULL
26	No Outcome	08-Apr-2020	32	No	Female	(E) Other Ethnic Groups - Prefer ...	No	Prefer not to say religion	Prefer not to say sexuality	ISCED Level 2 (Lower secondary e...	Yes
27	No Outcome	20-Mar-2021	40	NULL	Female	(D) Black or Black British - Other ...	NULL	Christian	Heterosexual	NULL	NULL
28	No Outcome	19-Aug-2020	40	Yes	Female	(E) Other Ethnic Groups - Any ot...	No	Muslim	Heterosexual	ISCED Level 5 (Short cycle tertiary ...	No
29	No Outcome	20-Jul-2021	47	NULL	Female	(A) White - Kurdish	NULL	Muslim	Heterosexual	NULL	NULL
30	No Outcome	11-Dec-2019	36	Yes	Female	(D) Black or Black British - Carib...	Yes	No religion / atheist	Heterosexual	ISCED Level 2 (Lower secondary e...	Yes
31	No Outcome	24-Mar-2021	24	NULL	Female	(E) Other Ethnic Group - Latin A...	NULL	Other religion	Heterosexual	NULL	NULL
32	No Outcome	05-Feb-2021	42	Yes	Female	(D) Black or Black British - Other ...	Yes	Muslim	Heterosexual	ISCED Level 2 (Lower secondary e...	Yes
33	No Outcome	23-Jun-2021	18	NULL	Male	(E) Other Ethnic Group - Latin A...	NULL	No religion / atheist	Heterosexual	NULL	NULL

As you can see all of the “Blanks” are now NULL.

I am now going to identify all the variables that have more than 50% NULL values and then I will do further analysis to see their importance on the target variable “Employer” and whether or not they can be removed or not.

Starting off with identifying variables with more than 50% missing values, I have done this by generating the frequency distribution tables of the remaining variables for each table I looked for the “NULL” value and looked at the percent column, if this is above 50% I will delete it.



The following variables are those with more than 50% missing values.

Parent_On_Enrolment – NULL = 55.8%

Parent_on_Enrolment

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	1087	22.7	22.7	22.7
	NULL	2673	55.8	55.8	78.5
	Yes	1028	21.5	21.5	100.0
	Total	4788	100.0	100.0	

Has_Disability – NULL = 52.9%

Has_Disability

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	1617	33.8	33.8	33.8
	NULL	2531	52.9	52.9	86.6
	Prefer not to say	67	1.4	1.4	88.0
	Yes	573	12.0	12.0	100.0
	Total	4788	100.0	100.0	

Highest_Level_Of_Education – NULL = 56.1%

Highest_Level_of_Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	ISCED Level 0 (Early childhood education)	49	1.0	1.0	1.0
	ISCED Level 1 (Primary education)	174	3.6	3.6	4.7
	ISCED Level 2 (Lower secondary education)	633	13.2	13.2	17.9
	ISCED Level 3 (Upper secondary education)	463	9.7	9.7	27.5
	ISCED Level 4 (Post secondary - tertiary and non-tertiary)	183	3.8	3.8	31.4
	ISCED Level 5 (Short cycle tertiary education)	96	2.0	2.0	33.4
	ISCED Level 6 (Bachelor's or equivalent level)	390	8.1	8.1	41.5
	ISCED Level 7 (Master's or equivalent level)	107	2.2	2.2	43.8
	ISCED Level 8 (Doctoral or equivalenmt level)	9	.2	.2	43.9
	NULL	2684	56.1	56.1	100.0
	Total	4788	100.0	100.0	

Claiming_Benefits – NULL = 55.4%

Claiming_Benefits

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	811	16.9	16.9	16.9
	NULL	2651	55.4	55.4	72.3
	Yes	1326	27.7	27.7	100.0
	Total	4788	100.0	100.0	

Benefits – NULL = 52.4%

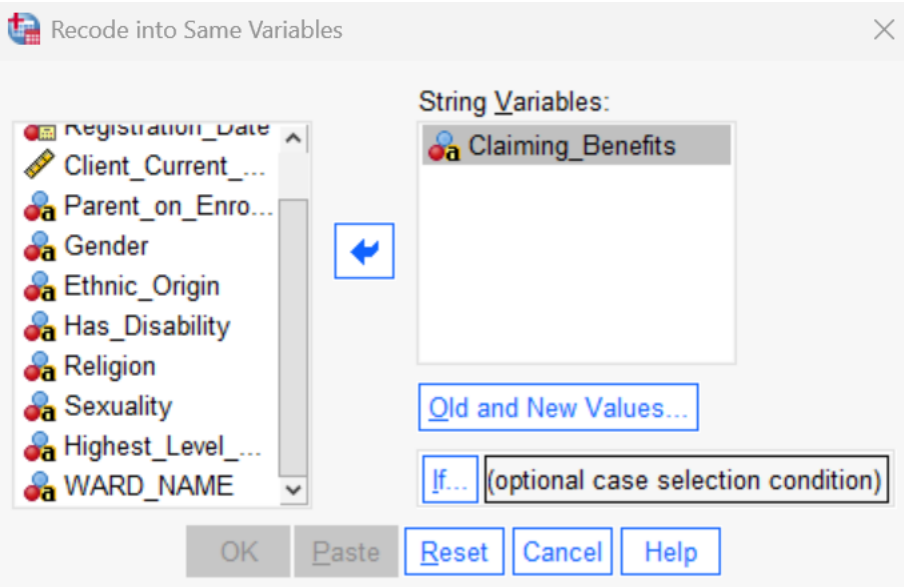
Job seekers allowance (JSA), Other benefit, Months on benefits	4	.1	.1	10.3
Job seekers allowance (JSA), Personal independence payment (PI)	2	.0	.0	10.4
Job seekers allowance (JSA), Universal credit (UC)	4	.1	.1	10.4
Job seekers allowance (JSA), Universal credit (UC), Months on benefits	1	.0	.0	10.5
Months on benefits	15	.3	.3	10.8
Months on benefits, Not in receipt of benefits	4	.1	.1	10.9
Not in receipt of benefits	783	16.4	16.4	27.2
NULL	2507	52.4	52.4	79.6
Other benefit	78	1.6	1.6	81.2
Other benefit, Months on benefits	10	.2	.2	81.4
Other benefit, Not in receipt of benefits	2	.0	.0	81.5
Other benefit, Working Tax Credit, Months on benefits	1	.0	.0	81.5
Personal independence payment (PI)	11	.2	.2	81.7
Personal independence payment (PI), Disability living allowance (DLA)	3	.1	.1	81.8

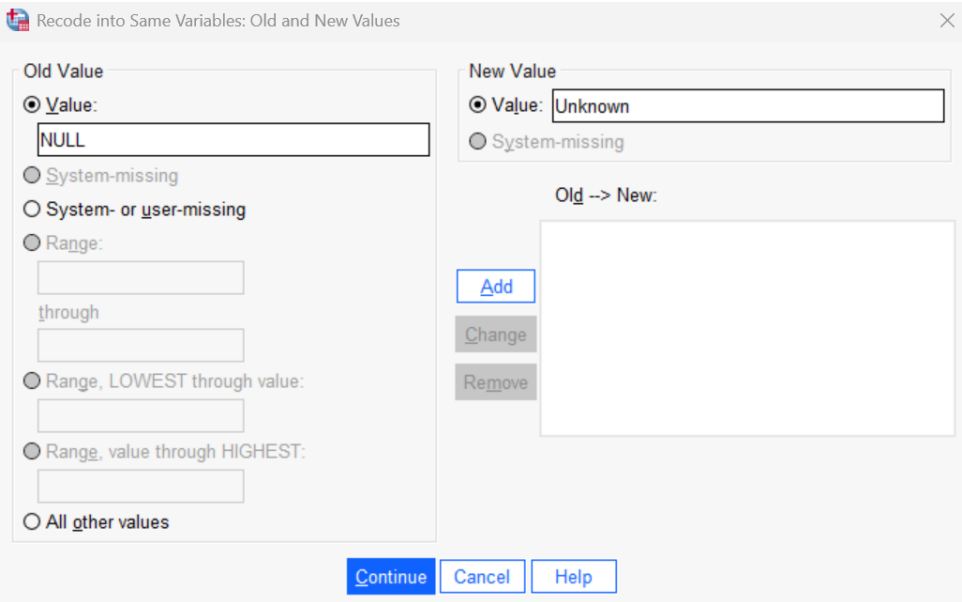
Even though these variables have more than 50% missing data, we can not just remove them based on that, we have to do further analysis to see if they are important on the target variable. I could carry out some analysis techniques such as a chi-square test however I don't feel like this is necessary as it doesn't understand the context so I will use my tuition on what to do with these variables.

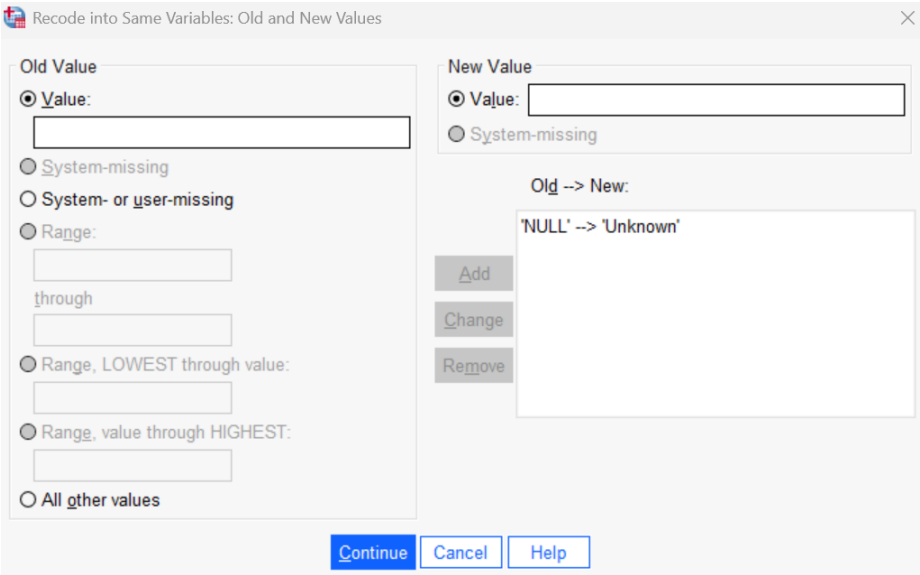
I have identified the following as critical variables, these are the variables that I feel cannot be removed from the data set, "Has_Disability", "Highest_Level_Of_Education" and "Claiming_Benefits". The variable "Benefits" can be removed as I don't feel this is necessary for the target variable as "Claiming_Benefits" already give us a binary yes or no if they are on benefits or not. The last variable to think about is "Parent_On_Enrolment", from my understanding this means if the person asked was a parent at the time of enrolment or not, I feel as if this variable should be kept as being a parent might pose barriers to being employed.

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	Employer	String	48	0		None	None	48	Left	Nominal	Input
2	Registration_Date	Date	11	0		None	None	10	Right	Nominal	Input
3	Client_Current_Age	Numeric	3	0		None	None	8	Right	Scale	Input
4	Parent_on_Enrolment	String	6	0		None	None	6	Left	Nominal	Input
5	Gender	String	17	0		None	None	17	Left	Nominal	Input
6	Ethnic_Origin	String	55	0		None	None	26	Left	Nominal	Input
7	Has_Disability	String	17	0		None	None	17	Left	Nominal	Input
8	Religion	String	26	0		None	None	26	Left	Nominal	Input
9	Sexuality	String	27	0		None	None	27	Left	Nominal	Input
10	Highest_Level_of_Education	String	58	0		None	None	28	Left	Nominal	Input
11	Claiming_Benefits	String	6	0		None	None	6	Left	Nominal	Input
12	WARD_NAME	String	31	0		None	None	31	Left	Nominal	Input

These are the remaining variables, now we have to decide what to do with the variables that have more than 50% missing values that I have decided not to remove, I will replace the “NULL” variable with some over value, lets begin with “Claiming_Benefits”. I have decided to recode the “NULL” values into “Unknown” as to avoid any bias of replacing values with either yes or no.





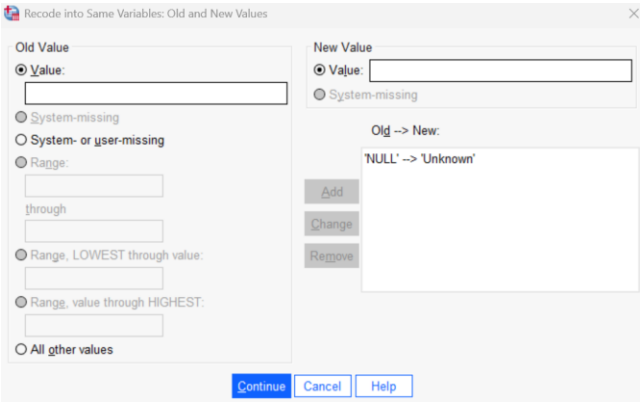


Claiming_Benefits

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	811	16.9	16.9	16.9
	Unknown	2651	55.4	55.4	72.3
	Yes	1326	27.7	27.7	100.0
	Total	4788	100.0	100.0	

This is the outcome of the “Claiming_Benefits” variable, “NULL” has been replaced by “Unknown”

I will now move on to the “Highest_Level_Of_Education” variable, I have done multiple chi square tests to test the significance and the p value shows that there is a significant correlation (< 0.05) with the other variables therefore I cant simply remove the variable and I also cant use mode imputation as it would introduce a bias. therefore I will have to do the same as what I did for “Claiming_Benefits” and recode the “NULL” into “Unknown”.



Highest_Level_of_Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	ISCED Level 0 (Early childhood education)	49	1.0	1.0	1.0
	ISCED Level 1 (Primary education)	174	3.6	3.6	4.7
	ISCED Level 2 (Lower secondary education)	633	13.2	13.2	17.9
	ISCED Level 3 (Upper secondary education)	463	9.7	9.7	27.5
	ISCED Level 4 (Post secondary - tertiary and non-tertiary)	183	3.8	3.8	31.4
	ISCED Level 5 (Short cycle tertiary education)	96	2.0	2.0	33.4
	ISCED Level 6 (Bachelor's or equivalent level)	390	8.1	8.1	41.5
	ISCED Level 7 (Master's or equivalent level)	107	2.2	2.2	43.8
	ISCED Level 8 (Doctoral or equivalenmt level)	9	.2	.2	43.9
	Unknown	2684	56.1	56.1	100.0
	Total	4788	100.0	100.0	

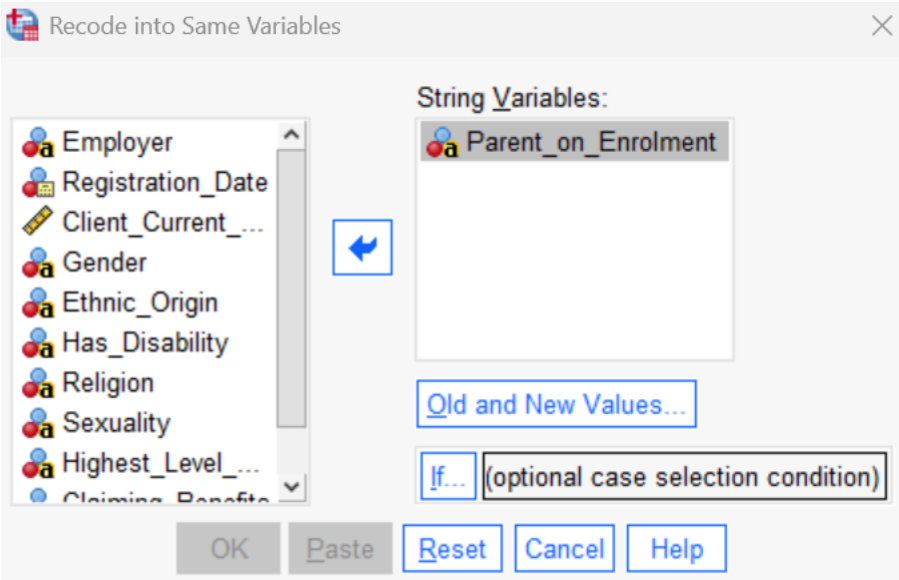
As you can see “NULL” is now “Unknown”.

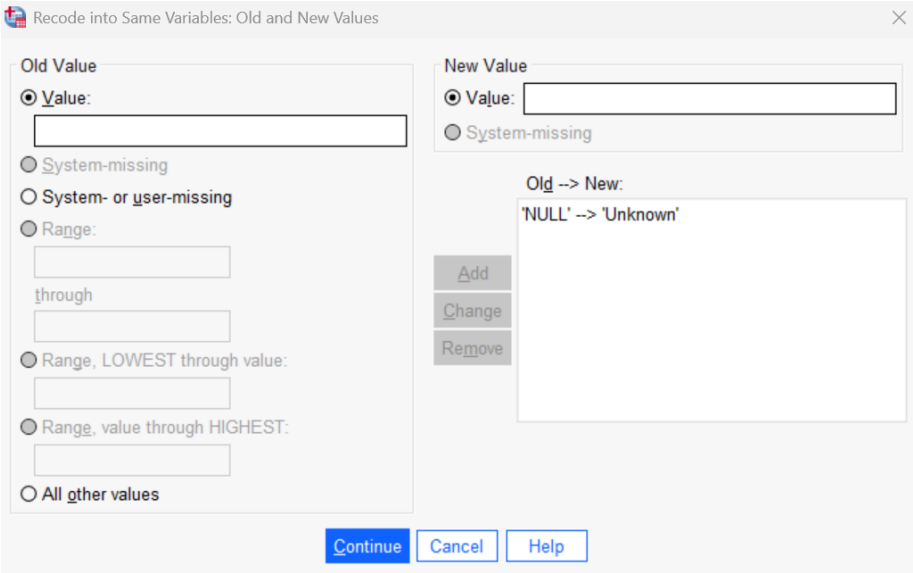
Lets move on to “Parent_On_Enrolment” as you can see from the frequency distribution table “yes” makes up for 21.5% and “no” makes up for 22.7% while “NULL” makes up for 55.8%

Parent_on_Enrolment

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Blanks	2673	55.8	55.8	55.8
	No	1087	22.7	22.7	78.5
	Yes	1028	21.5	21.5	100.0
	Total	4788	100.0	100.0	

I feel giving this coursework context it would be inappropriate to just delete the whole column so I will replace the “NULL” values with “Unknown”.





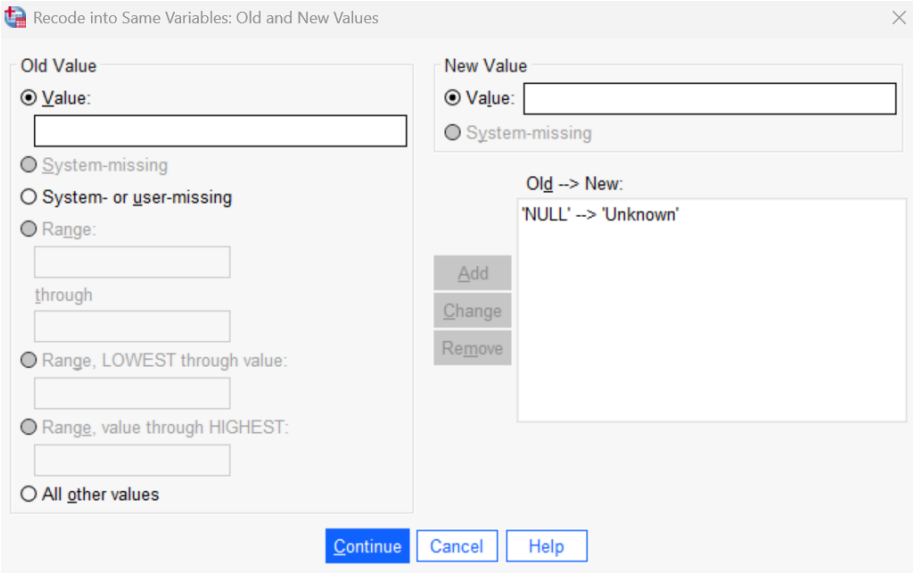
Parent_on_Enrolment

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	1087	22.7	22.7	22.7
	Unknown	2673	55.8	55.8	78.5
	Yes	1028	21.5	21.5	100.0
	Total	4788	100.0	100.0	

Lets move on to the last variable that is “Has_Disability” as you can see from the frequency distribution table “NULL” makes up for 52.9% of the values in the variable, I will have to change “NULL” to “Unknown” as given the context having a disability will be very significant.

Has_Disability

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	1617	33.8	33.8	33.8
	NULL	2531	52.9	52.9	86.6
	Prefer not to say	67	1.4	1.4	88.0
	Yes	573	12.0	12.0	100.0
	Total	4788	100.0	100.0	



Has_Disability

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	1617	33.8	33.8	33.8
	Prefer not to say	67	1.4	1.4	35.2
	Unknown	2531	52.9	52.9	88.0
	Yes	573	12.0	12.0	100.0
	Total	4788	100.0	100.0	

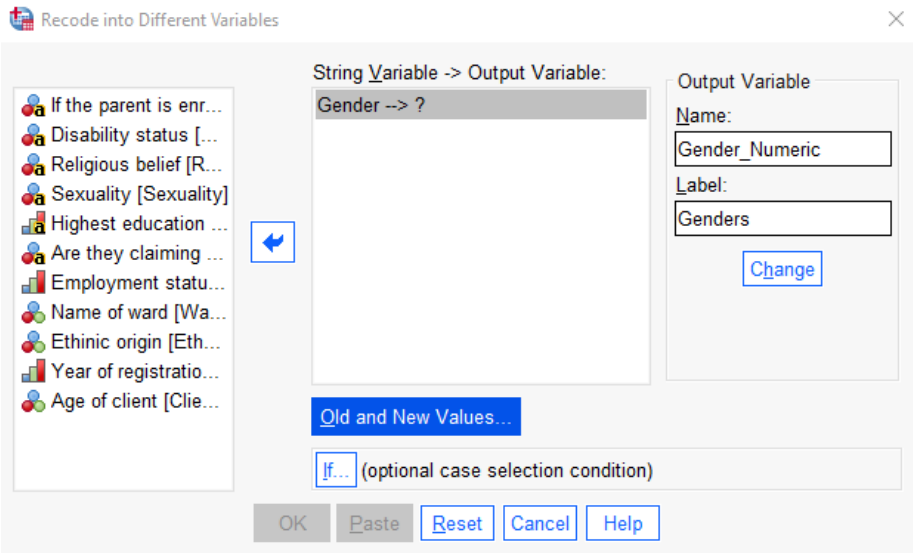
As you can see the changes have been made.

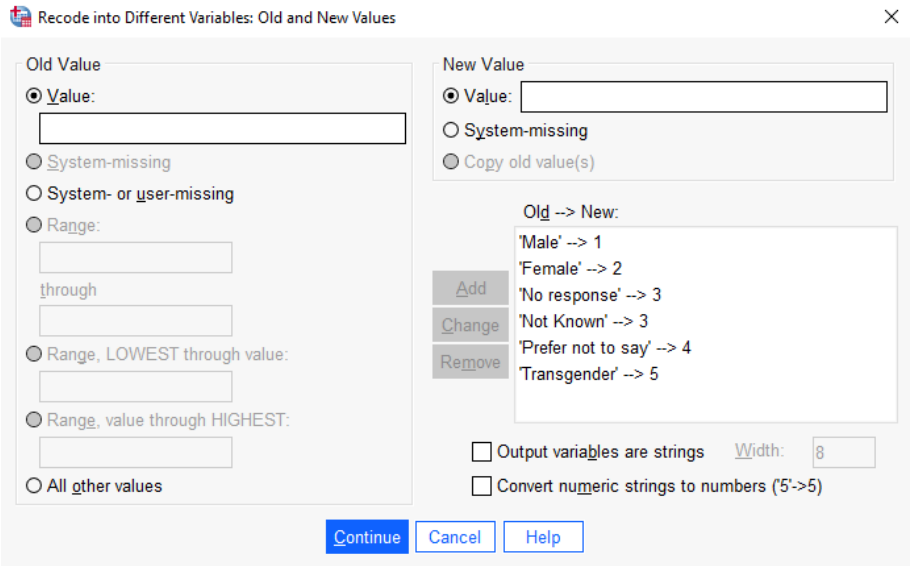
Now that we have dealt with the missing values from the variables I will move onto preparing data in numerical where it is possible, first I will identify which variables would be suitable to convert.

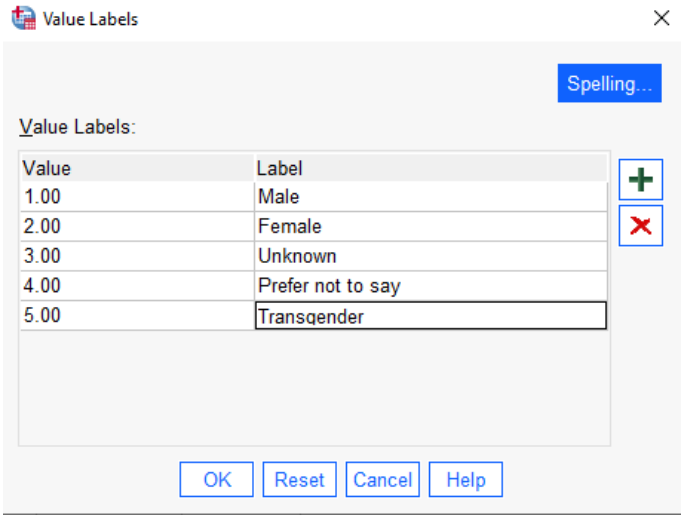
I will start with the most obvious one “Gender”

Gender

I will transform this into a new numerical variable.

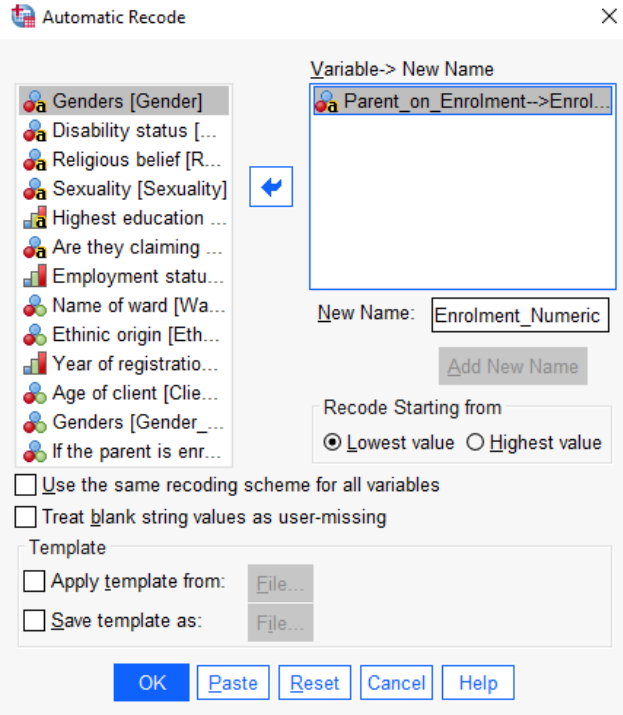


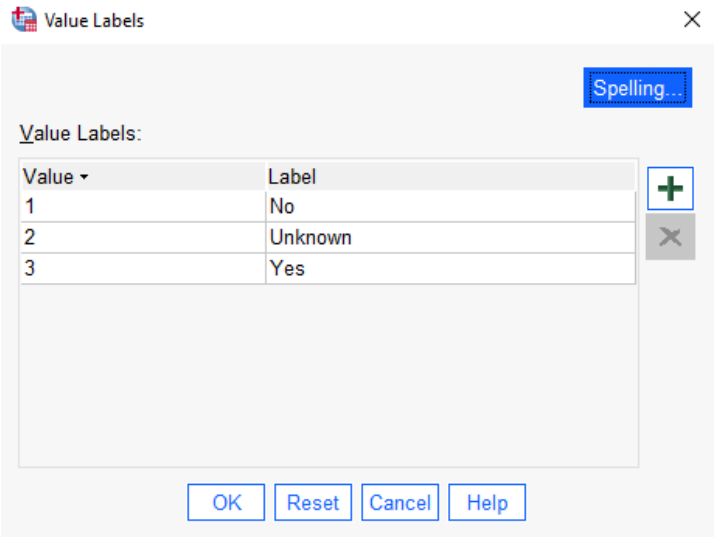




Parent_on_Enrolment

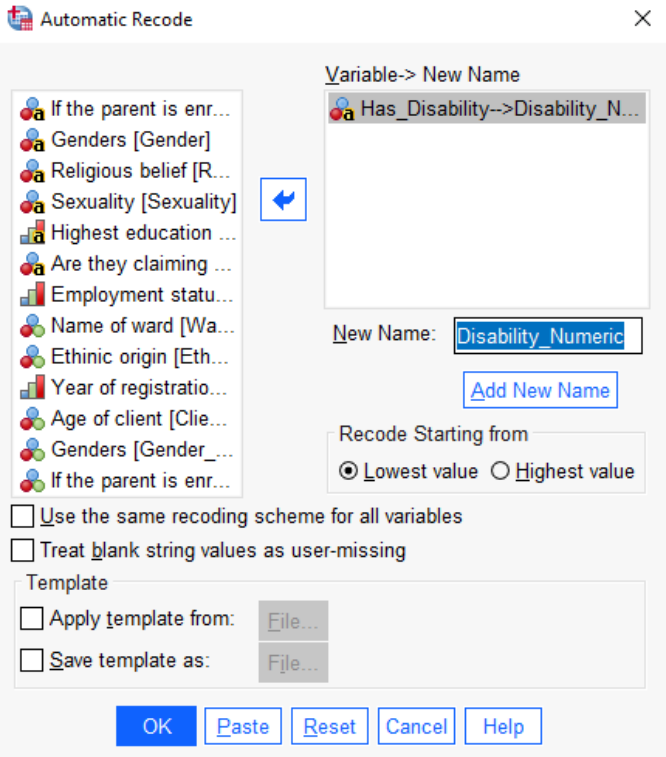
I will use automatic recoding as I don't have to change any values.

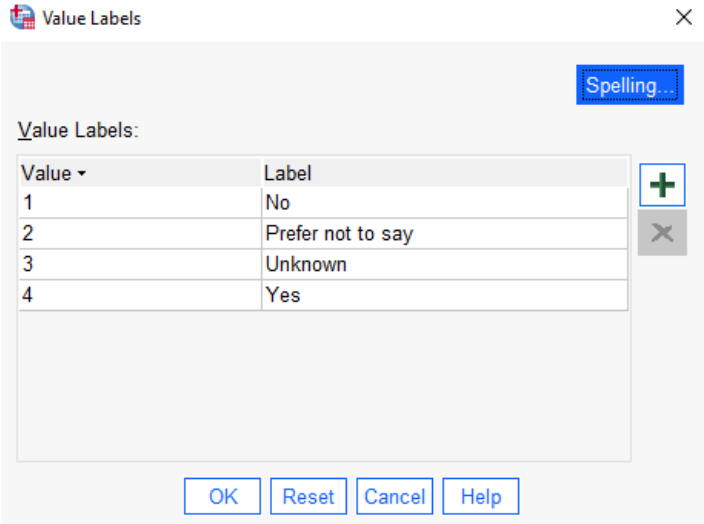




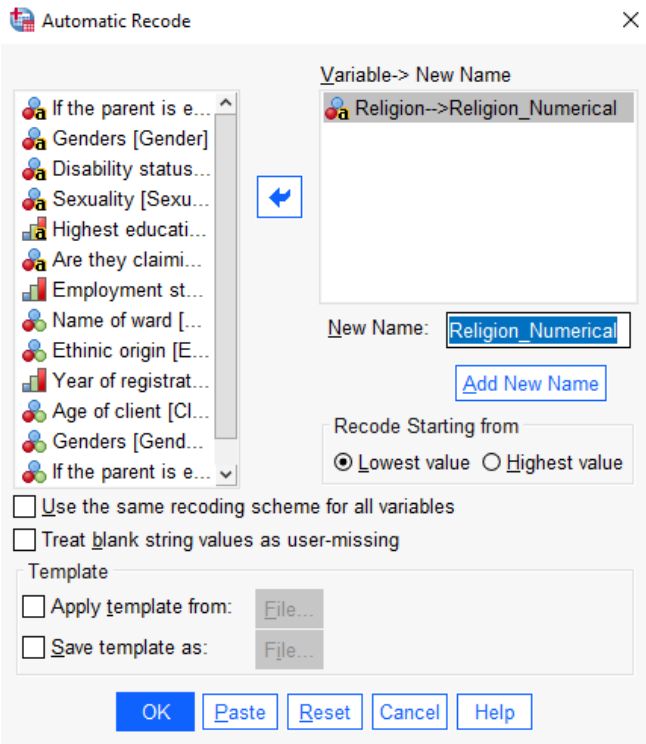
Has_Disability

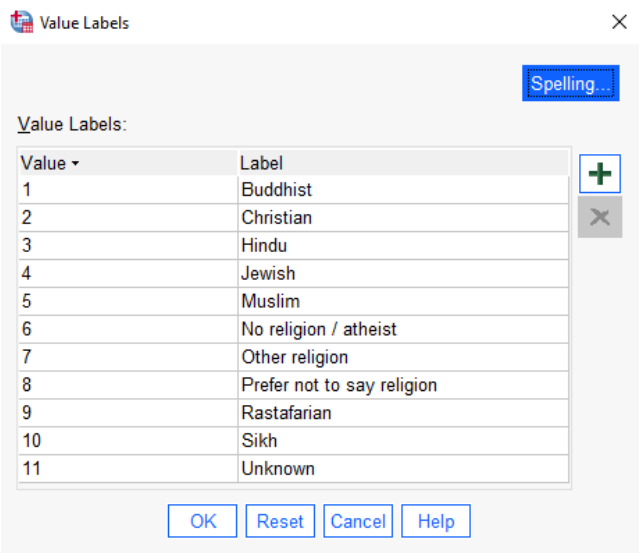
I will use automatic recoding as I don't have to change any values.



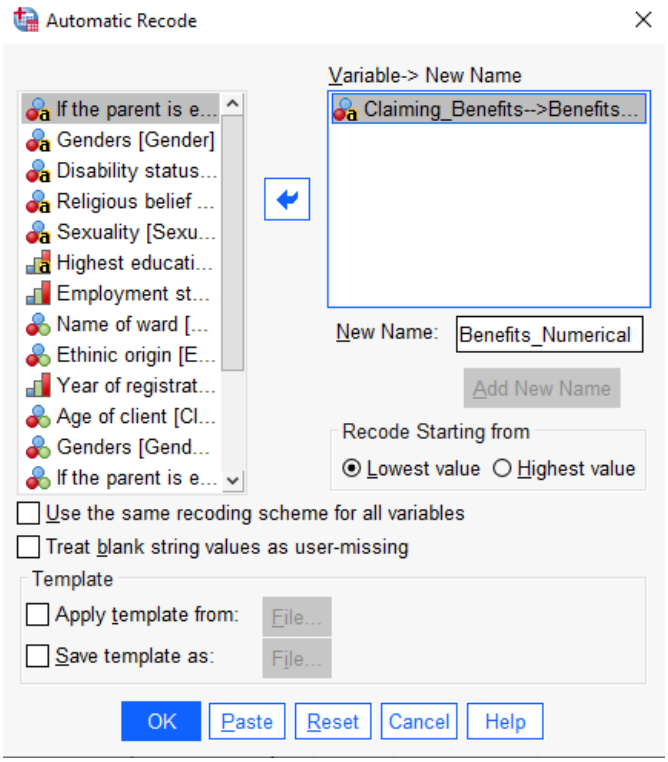


Religion



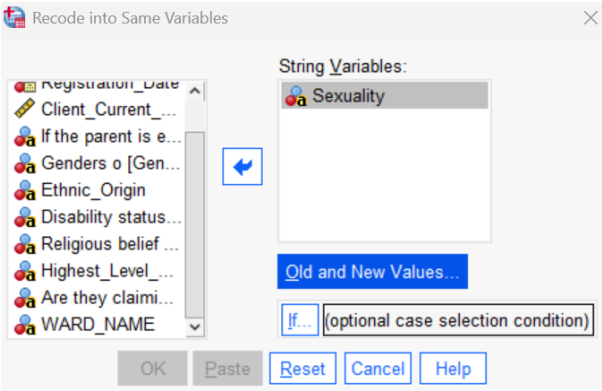


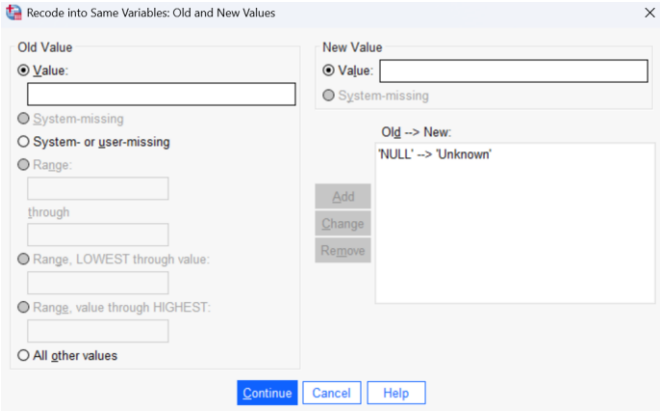
Claiming_Benefits



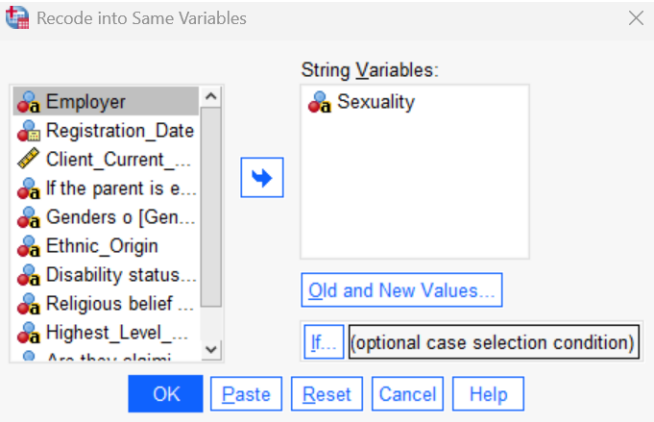
Sexuality

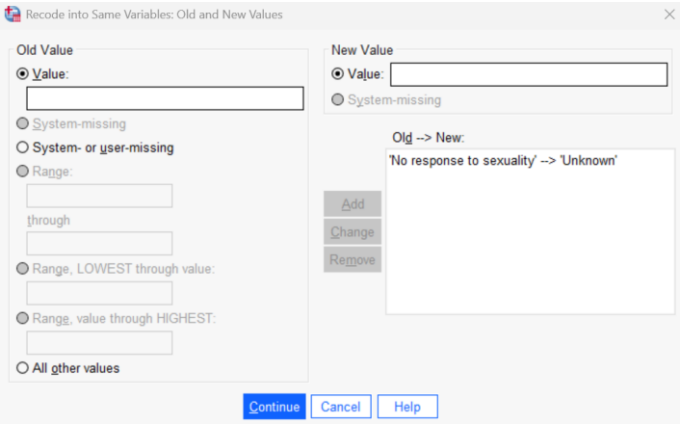
Firstly I need to transform the “NULL” variables.

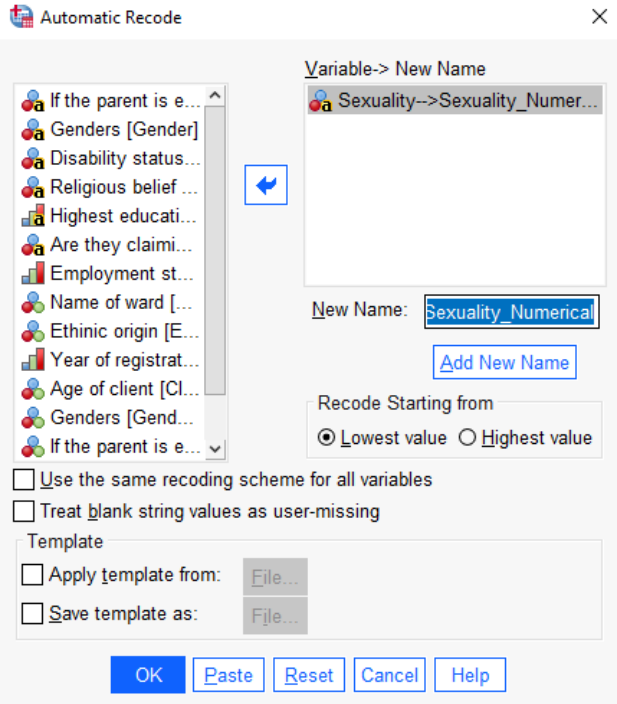




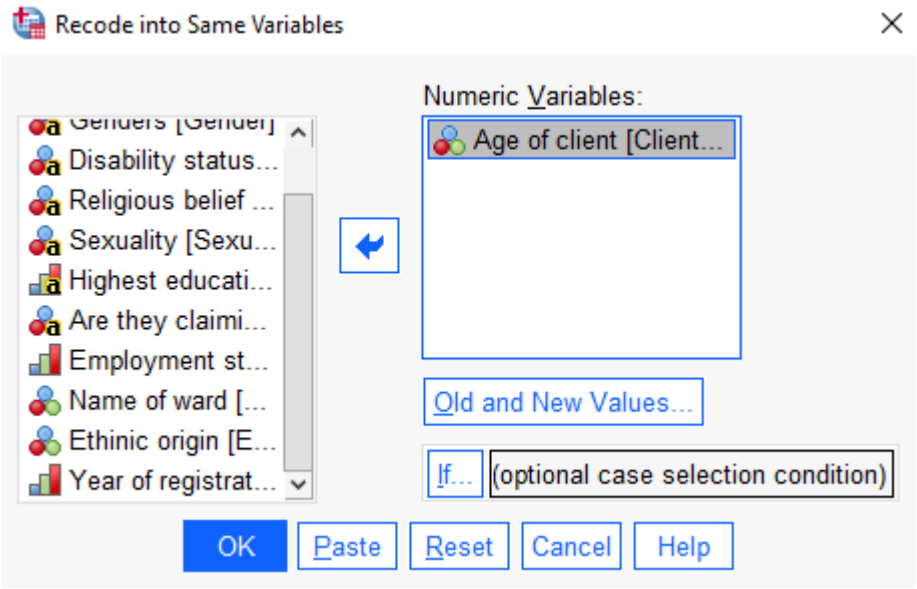
Again I will change “No response to sexuality” to “Unknown”.

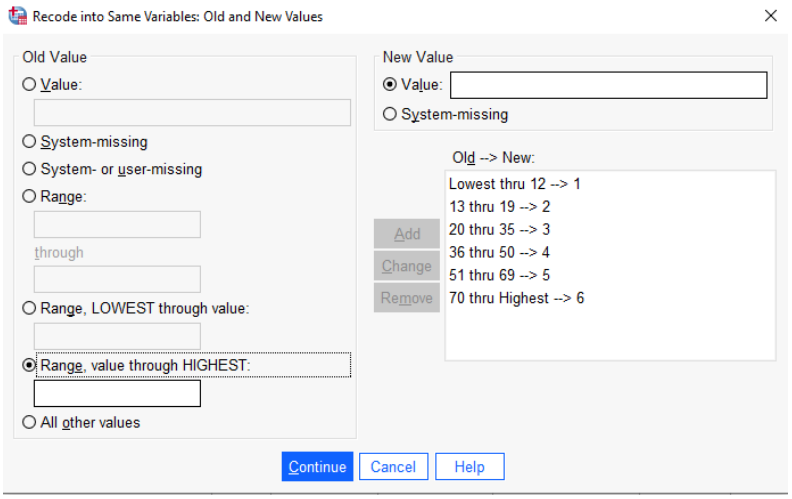






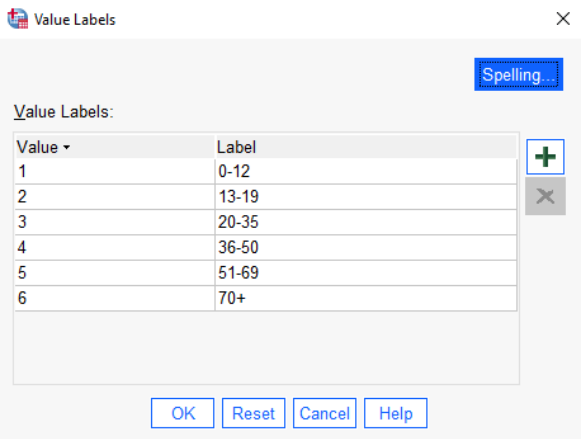
Client_Current_Age





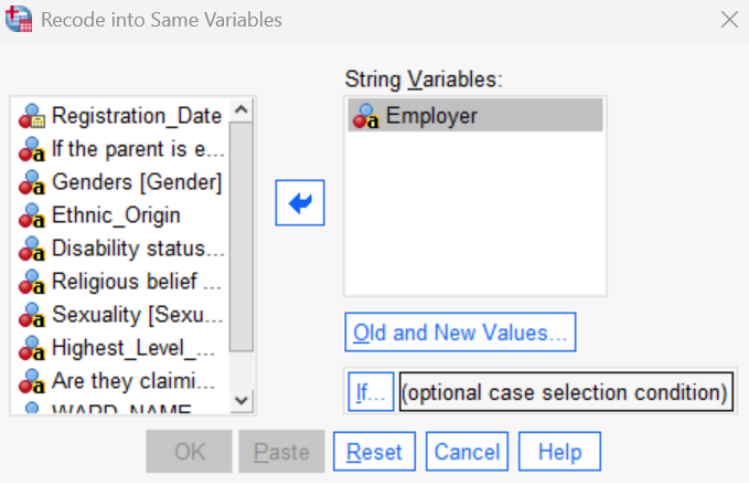
As you can see I have used the range function to define the age groups. My reasoning is as follows.

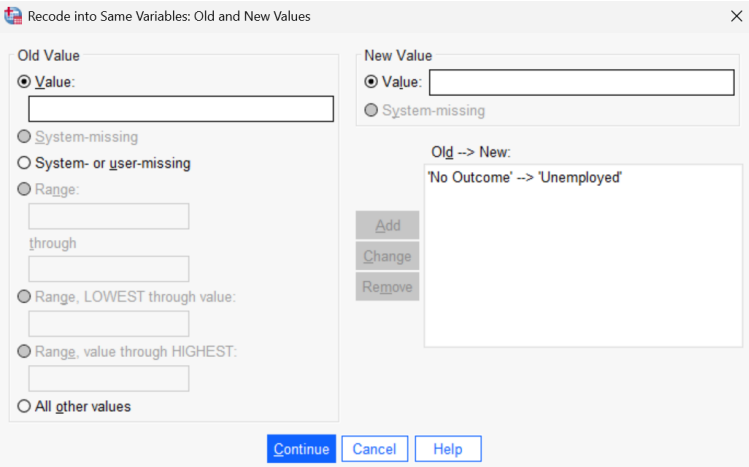
0 – 12 are children
13 – 19 are teenagers
20 – 35 are young adults
36 – 50 are middle aged adults
51 – 69 are older adults
70 thu highest are elderly



Employer

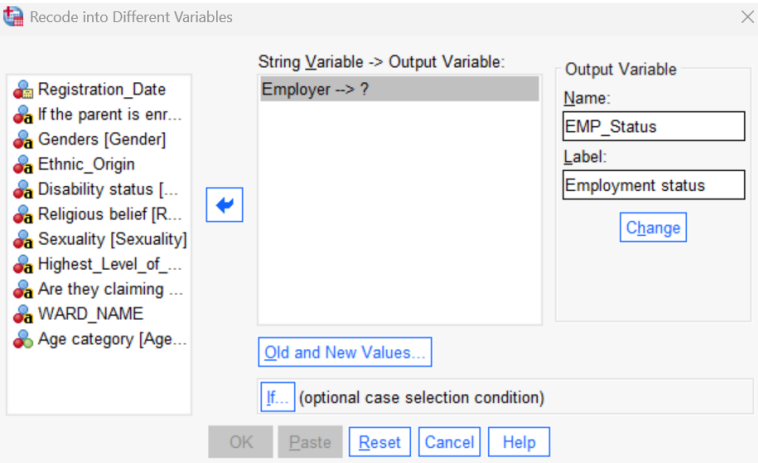
As this is the target variable we have to take extra care not to introduce bias or any other factor that might change the results, 87.4% of the responses have “No outcome”, given the project parameters I can assume that this means that they were not employed otherwise there would have been an employer name. I will change “No Outcome” to “Unemployed” for better understanding of the data.

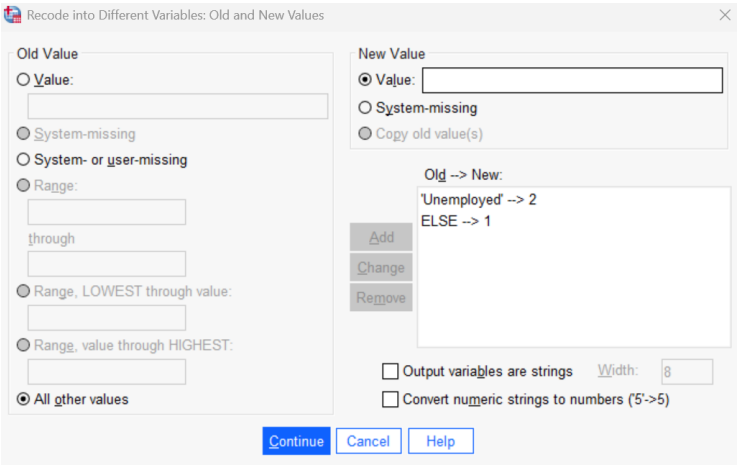




Unemployed	4184	87.4	87.4	99.6

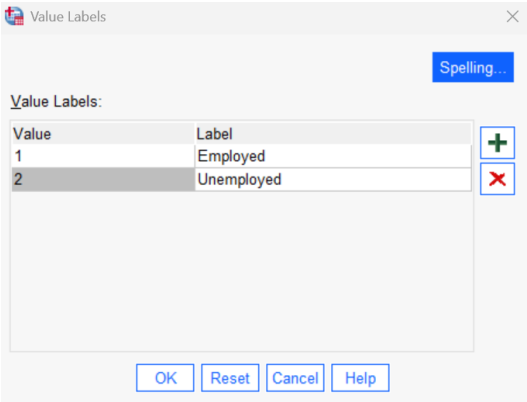
Now that is done I will now group the data into two different categories being “Employed” and “Unemployed” as the name of the employer isn’t important only if they have been employed or not.



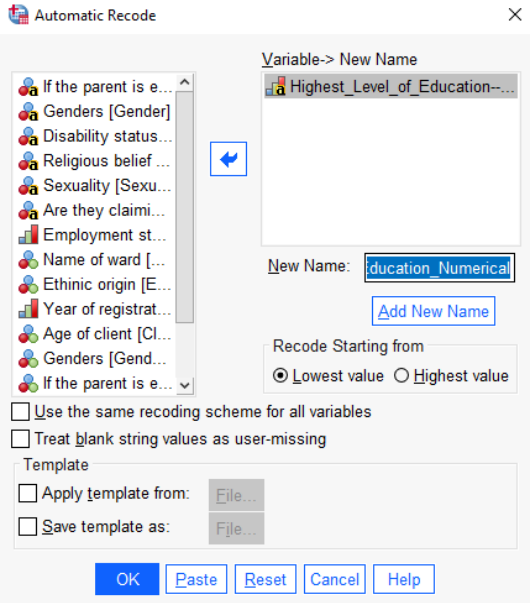


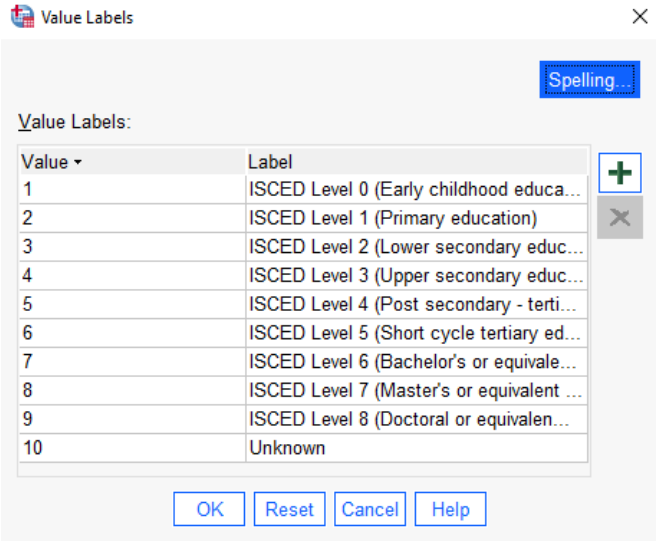
12	EMP_Status	Numeric	8	0	Employment status	{1, Employe...	None	12	Right	Ordinal	Input

Now there is a new variable called “EMP_Status” which is ordinal as it has a ranking, I will remove the original variable and add values as follows, “Employed” as 1 and “Unemployed” as 2.



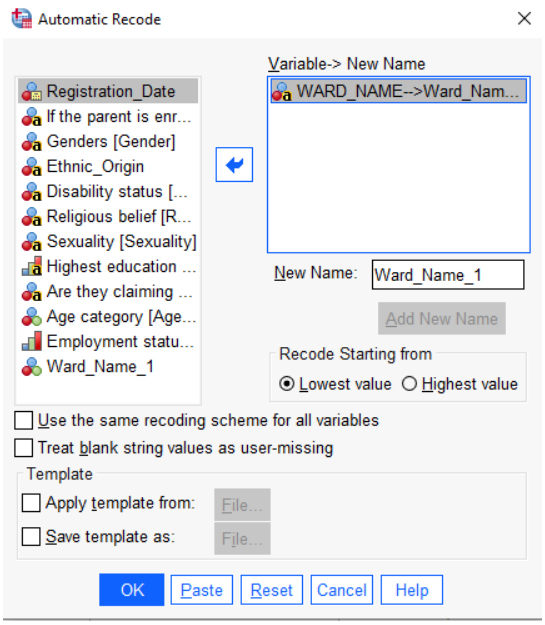
Highest_Level_Of_Education





WARD_NAME

As this variable has a very large amount of different outcomes, I have used the automatic recode function to add value labels.

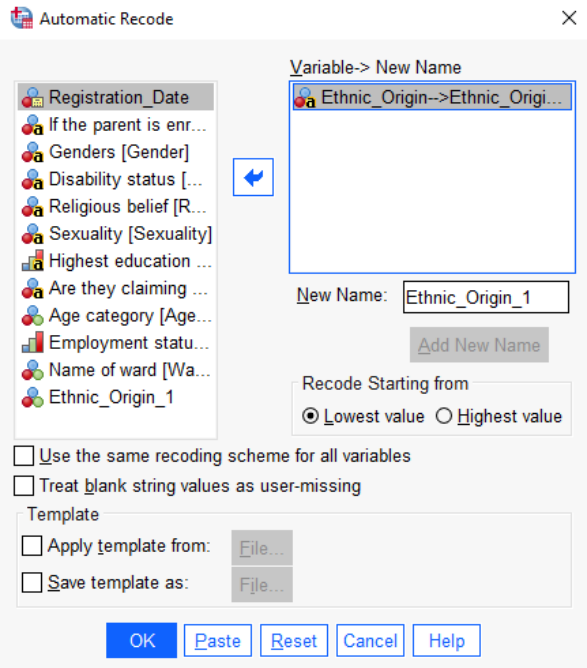


I will delete the original variable.

12	Ward_Nam...	Numeric	3	0	Name of ward	{1, Abbey R...	None	13	Right	Nominal	Input

It has been labelled.

Ethnic_Origin

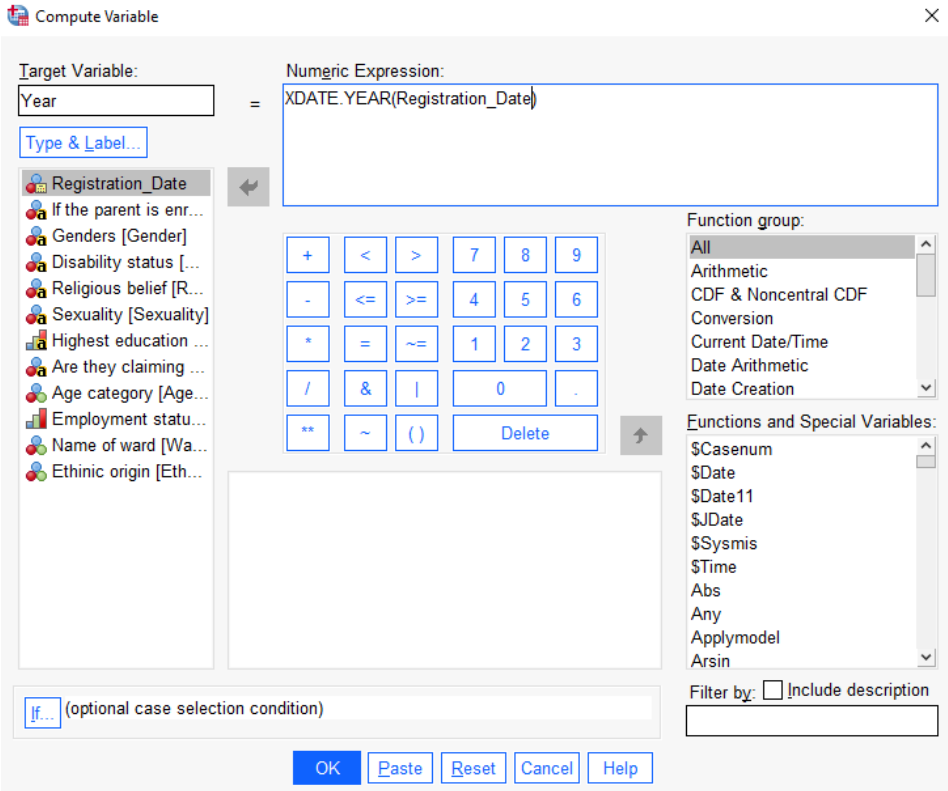


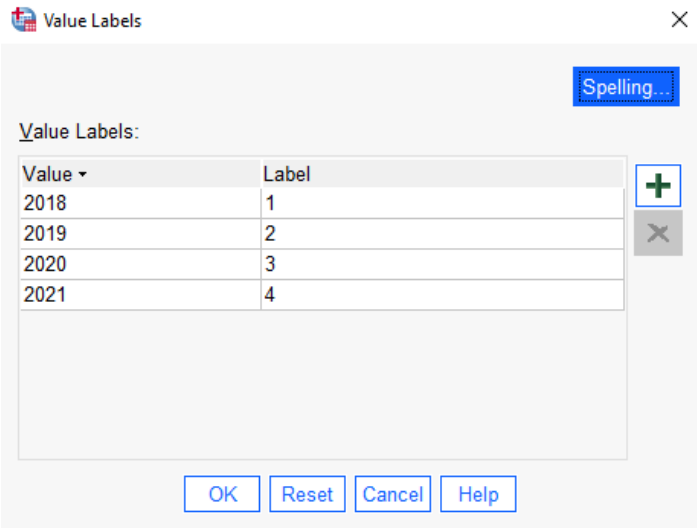
12	Ethnic_Origin	Numeric	2	0	Ethinic origin	{1, (A) White...	None	17	Right	Nominal	Input

Variable has been renamed and labelled.

Registration_Date

I need to group the outcomes by the year so I will do this by using this expression.





I have tidied the data by removing the original data and renaming the variables just created.

	EMP_Status	Ward_Name	Ethnic_Origin	Year	Client_Current_Age	Gender	Parent_On_Enrolment	Has_Disability	Religion	Sexuality	HIghest_Level_Of_Education	Claiming_Benefits
1	2	199	12	2021	3	2	2	1	11	5	7	1
2	2	172	12	2021	3	1	2	1	11	5	7	1
3	2	64	22	2021	4	2	2	3	2	3	10	2
4	2	138	17	2021	5	1	2	1	11	6	8	3
5	2	9	1	2021	3	2	2	1	11	6	3	3
6	2	105	2	2021	5	1	2	1	11	6	7	3
7	2	105	15	2021	3	1	2	1	11	6	8	1
8	2	127	12	2021	3	1	2	1	11	5	4	3
9	2	127	13	2021	2	2	2	4	11	5	4	1
10	2	28	18	2021	3	1	2	2	11	5	10	1
11	2	89	2	2021	3	2	2	3	6	3	10	2
12	2	89	13	2021	3	2	2	3	5	3	10	2
13	2	12	2	2021	3	2	2	3	6	3	10	2
14	2	111	17	2021	3	2	2	1	11	6	5	1
15	2	115	2	2021	4	2	2	3	6	3	10	2
16	1	65	2	2020	4	2	1	1	8	5	7	1
17	2	120	17	2021	3	2	2	1	11	6	8	1
18	2	60	7	2021	4	2	2	3	5	3	10	2
19	2	36	1	2019	4	1	1	1	6	3	4	3
20	2	36	18	2021	4	2	2	3	8	3	10	2
21	2	23	2	2019	3	2	3	1	6	3	5	3
22	2	23	23	2021	4	2	2	3	5	3	10	2
23	2	23	12	2021	4	1	2	1	11	5	8	1
24	2	23	18	2021	4	1	2	3	8	1	10	2
25	2	23	10	2021	4	1	2	3	2	1	10	2
26	2	23	29	2020	3	2	1	1	8	5	3	3
27	2	23	22	2021	4	2	2	3	2	3	10	2
28	2	23	26	2020	4	2	3	1	5	3	6	1
29	2	23	6	2021	4	2	2	3	5	3	10	2
30	2	23	18	2019	4	2	3	4	6	3	3	3
31	2	23	25	2021	3	2	2	3	7	3	10	2
32	2	23	22	2021	4	2	3	4	5	3	3	3
33	2	23	25	2021	2	1	2	3	6	3	10	2
34	1	23	22	2021	5	1	3	1	5	3	7	2
35	2	23	11	2021	3	2	2	3	2	3	10	2
36	2	170	2	2021	3	1	2	3	6	3	10	2
37	2	170	2	2020	3	1	3	1	6	3	3	3
38	2	23	12	2021	4	4	2	3	8	5	10	2
39	2	23	1	2021	5	2	2	3	8	3	10	2

	EMP_Status	Ward_Name	Ethnic_Origin	Year	Client_Current_Age	Gender	Parent_On_Enrolment	Has_Disability	Religion	Sexuality	HIghest_Level_Of_Education	Claiming_Benefits
1	Unemployed	Whitechapel	(C) Asian or Asian Br...	4	20-35	Female	Unknown	No	Unknown	Prefer not to say sexuality	ISCED Level 6 (Bachelor'...	No
2	Unemployed	Stepney Green	(C) Asian or Asian Br...	4	20-35	Male	Unknown	No	Unknown	Prefer not to say sexuality	ISCED Level 6 (Bachelor'...	No
3	Unemployed	Forest	(D) Black or Black Bri...	4	36-50	Female	Unknown	Unknown	Christian	Heterosexual	Unknown	Unknown
4	Unemployed	NULL	(D) Black or Black Bri...	4	51-69	Male	Unknown	No	Unknown	Unknown	ISCED Level 7 (Master's o...	Yes
5	Unemployed	Blackwall & Cu...	(A) White - Any other ...	4	20-35	Female	Unknown	No	Unknown	Unknown	ISCED Level 2 (Lower sec...	Yes
6	Unemployed	Island Gardens	(A) White - British	4	51-69	Male	Unknown	No	Unknown	Unknown	ISCED Level 6 (Bachelor'...	Yes
7	Unemployed	Island Gardens	(C) Asian or Asian Br...	4	20-35	Male	Unknown	No	Unknown	Unknown	ISCED Level 7 (Master's o...	No
8	Unemployed	Mile End	(C) Asian or Asian Br...	4	20-35	Male	Unknown	No	Unknown	Prefer not to say sexuality	ISCED Level 3 (Upper sec...	Yes
9	Unemployed	Mile End	(C) Asian or Asian Br...	4	13-19	Female	Unknown	Yes	Unknown	Prefer not to say sexuality	ISCED Level 3 (Upper sec...	No
10	Unemployed	Canary Wharf	(D) Black or Black Bri...	4	20-35	Male	Unknown	Prefer not to say	Unknown	Prefer not to say sexuality	Unknown	No
11	Unemployed	High Street	(A) White - British	4	20-35	Female	Unknown	Unknown	No religion / atheist	Heterosexual	Unknown	Unknown
12	Unemployed	High Street	(C) Asian or Asian Br...	4	20-35	Female	Unknown	Unknown	Muslim	Heterosexual	Unknown	Unknown
13	Unemployed	Bow East	(A) White - British	4	20-35	Female	Unknown	Unknown	No religion / atheist	Heterosexual	Unknown	Unknown
14	Unemployed	King's Park	(D) Black or Black Bri...	4	20-35	Female	Unknown	No	Unknown	Unknown	ISCED Level 4 (Post seco...	No
15	Unemployed	Lea Bridge	(A) White - British	4	36-50	Female	Unknown	Unknown	No religion / atheist	Heterosexual	Unknown	Unknown
16	Employed	Forest Gate So...	(A) White - British	3	36-50	Female	No	No	Prefer not to say religion	Prefer not to say sexuality	ISCED Level 6 (Bachelor'...	No
17	Unemployed	London Fields	(D) Black or Black Bri...	4	20-35	Female	Unknown	No	Unknown	Unknown	ISCED Level 7 (Master's o...	No
18	Unemployed	Farringdon With...	(A) White - Turkish/ T...	4	36-50	Female	Unknown	Unknown	Muslim	Heterosexual	Unknown	Unknown
19	Unemployed	Clerkenwell	(A) White - Any other ...	2	36-50	Male	No	No	No religion / atheist	Heterosexual	ISCED Level 3 (Upper sec...	Yes
20	Unemployed	Clerkenwell	(D) Black or Black Bri...	4	36-50	Female	Unknown	Unknown	Prefer not to say religion	Heterosexual	Unknown	Unknown
21	Unemployed	Bunhill	(A) White - British	2	20-35	Female	Yes	No	No religion / atheist	Heterosexual	ISCED Level 4 (Post seco...	Yes
22	Unemployed	Bunhill	(D) Black or Black Bri...	4	36-50	Female	Unknown	Unknown	Muslim	Heterosexual	Unknown	Unknown
23	Unemployed	Bunhill	(C) Asian or Asian Br...	4	36-50	Male	Unknown	No	Unknown	Prefer not to say sexuality	ISCED Level 7 (Master's o...	No
24	Unemployed	Bunhill	(D) Black or Black Bri...	4	36-50	Male	Unknown	Unknown	Prefer not to say religion	Bisexual	Unknown	Unknown
25	Unemployed	Bunhill	(B) Mixed - White and...	4	36-50	Male	Unknown	Unknown	Christian	Bisexual	Unknown	Unknown
26	Unemployed	Bunhill	(E) Other Ethnic Gro...	3	20-35	Female	No	No	Prefer not to say religion	Prefer not to say sexuality	ISCED Level 2 (Lower sec...	Yes
27	Unemployed	Bunhill	(D) Black or Black Bri...	4	36-50	Female	Unknown	Unknown	Christian	Heterosexual	Unknown	Unknown
28	Unemployed	Bunhill	(E) Other Ethnic Gro...	3	36-50	Female	Yes	No	Muslim	Heterosexual	ISCED Level 5 (Short cycl...	No
29	Unemployed	Bunhill	(A) White - Kurdish	4	36-50	Female	Unknown	Unknown	Muslim	Heterosexual	Unknown	Unknown
30	Unemployed	Bunhill	(D) Black or Black Bri...	2	36-50	Female	Yes	Yes	No religion / atheist	Heterosexual	ISCED Level 2 (Lower sec...	Yes
31	Unemployed	Bunhill	(E) Other Ethnic Gro...	4	20-35	Female	Unknown	Unknown	Other religion	Heterosexual	Unknown	Unknown
32	Unemployed	Bunhill	(D) Black or Black Bri...	4	36-50	Female	Yes	Yes	Muslim	Heterosexual	ISCED Level 2 (Lower sec...	Yes
33	Unemployed	Bunhill	(E) Other Ethnic Gro...	4	13-19	Male	Unknown	Unknown	No religion / atheist	Heterosexual	Unknown	Unknown
34	Employed	Bunhill	(D) Black or Black Bri...	4	51-69	Male	Yes	No	Muslim	Heterosexual	ISCED Level 6 (Bachelor'...	Unknown
35	Unemployed	Bunhill	(B) Mixed - White and...	4	20-35	Female	Unknown	Unknown	Christian	Heterosexual	Unknown	Unknown
36	Unemployed	St Peter's	(A) White - British	4	20-35	Male	Unknown	Unknown	No religion / atheist	Heterosexual	Unknown	Unknown
37	Unemployed	St Peter's	(A) White - British	3	20-35	Male	Yes	No	No religion / atheist	Heterosexual	ISCED Level 2 (Lower sec...	Yes
38	Unemployed	Bunhill	(C) Asian or Asian Br...	4	36-50	Prefer not to say	Unknown	Unknown	Prefer not to say religion	Prefer not to say sexuality	Unknown	Unknown
39	Unemployed	Bunhill	(A) White - Any other ...	4	51-69	Female	Unknown	Unknown	Prefer not to say religion	Heterosexual	Unknown	Unknown

Checking and processing for out-of-range errors and outliers

Client_Current_Age

The minimum age for employment is 13 therefore we can remove the 0-12 group.

```
SELECT IF Client_Current_Age <> 1.
EXECUTE.
```

Age of client

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	13-19	275	5.8	5.8	5.8
	20-35	2059	43.2	43.2	49.0
	36-50	1520	31.9	31.9	80.9
	51-69	881	18.5	18.5	99.4
	70+	30	.6	.6	100.0
	Total	4765	100.0	100.0	

The average age of retirement is 65 therefore I will remove 70+

```
SELECT IF Client_Current_Age <> 6
EXECUTE.
```

Age of client

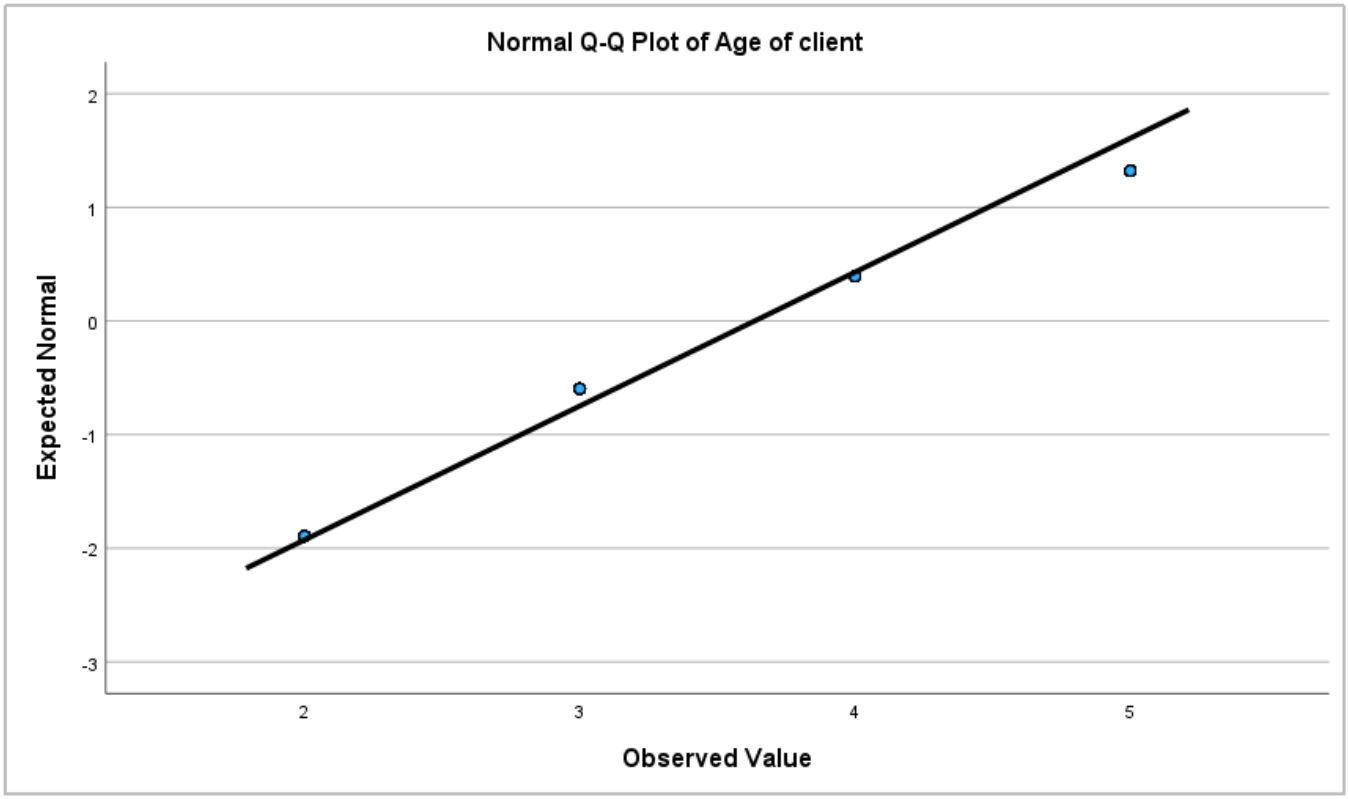
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	13-19	275	5.8	5.8	5.8
	20-35	2059	43.5	43.5	49.3
	36-50	1520	32.1	32.1	81.4
	51-69	881	18.6	18.6	100.0
	Total	4735	100.0	100.0	

I have not identified any other out of range errors or outliers as I have properly cleaned the data in previous steps.

Apply the Q-Q plot for normality analysis of the dataset

The only suitable variable for a Q-Q plot analysis is “Client_Current_Age” as it is the only continuous variable.

Client_Current_Age

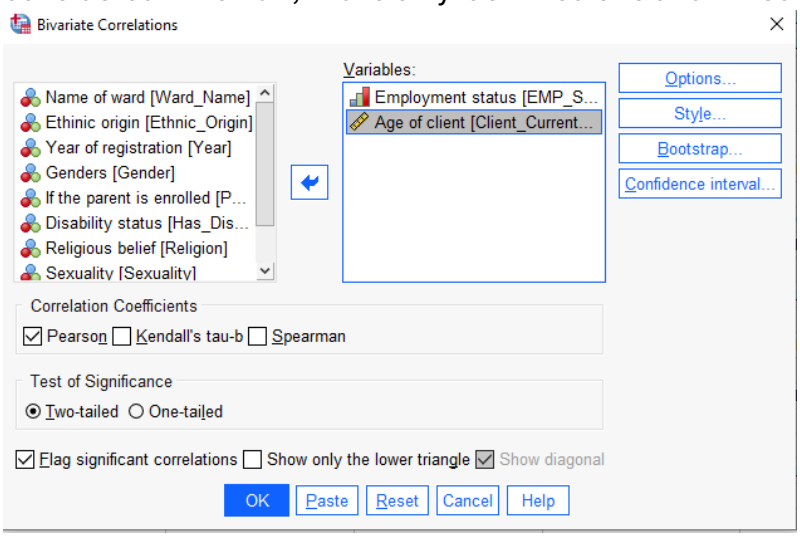


As you can see the points do follow the general line but has some deviation, especially with a large data set, however this does give us some evidence that the has normal distribution and is valid.

4. Correlation analysis

Correlation analysis of interval variables with the target variable

To show the correlation of the interval variables I must first identify which variables can be considered “interval”, I have only identified one and it would be “Client_Current_Age”.



Correlations

		Employment status	Age of client
Employment status	Pearson Correlation	1	.040**
	Sig. (2-tailed)		.006
	N	4735	4735
Age of client	Pearson Correlation	.040**	1
	Sig. (2-tailed)	.006	
	N	4735	4735

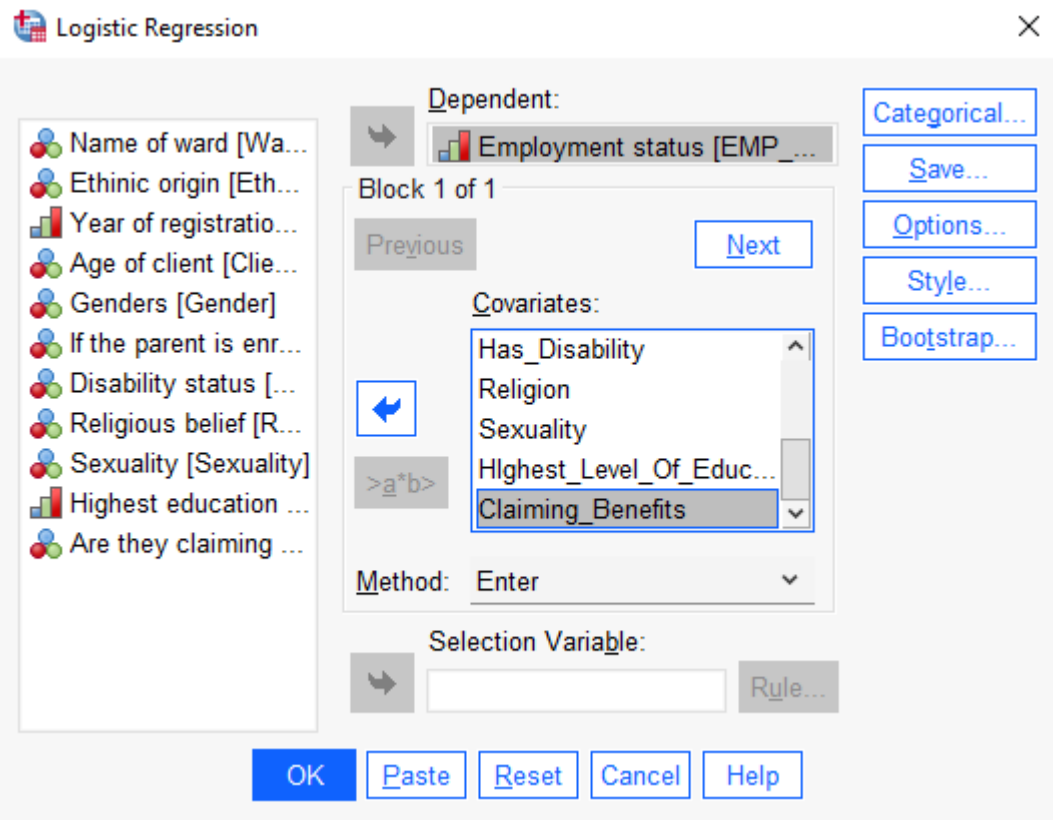
**. Correlation is significant at the 0.01 level (2-tailed).

Discussion and interpretation of the results

As we can see from the table I have used Pearson Correlation, the p value (Sig) being .006 gives us evidence that there is statistical significance between "Client_Current_Age" and the target variable "EMP_Status", the Pearson Correlation is .040 which indicates a very weak but still an increase in "Client_Current_Age" also increases "EMP_Status" this could mean that as the client gets older their employability increases.

5. Supervised learning model-based analysis

Apply logistic regression analysis



Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1[a]	Name of ward	.000	.001	.219	1	.640	1.000
	Ethinic origin	.002	.005	.098	1	.754	1.002
	Year of registration	.784	.050	249.456	1	<.001	2.189
	Genders	.099	.098	1.038	1	.308	1.105
	If the parent is enrolled	.224	.060	13.911	1	<.001	1.251
	Disability status	.282	.042	43.936	1	<.001	1.325
	Religious belief	-.010	.020	.259	1	.611	.990
	Sexuality	.060	.050	1.466	1	.226	1.062
	Highest education level	.106	.019	30.155	1	<.001	1.112
	Are they claiming benefits	.063	.060	1.083	1	.298	1.065
	Constant	-1582.983	100.135	249.907	1	<.001	.000

a. Variable(s) entered on step 1: Name of ward, Ethinic origin, Year of registration, Genders, If the parent is enrolled, Disability status, Religious belief, Sexuality, Highest education level, Are they claiming benefits.

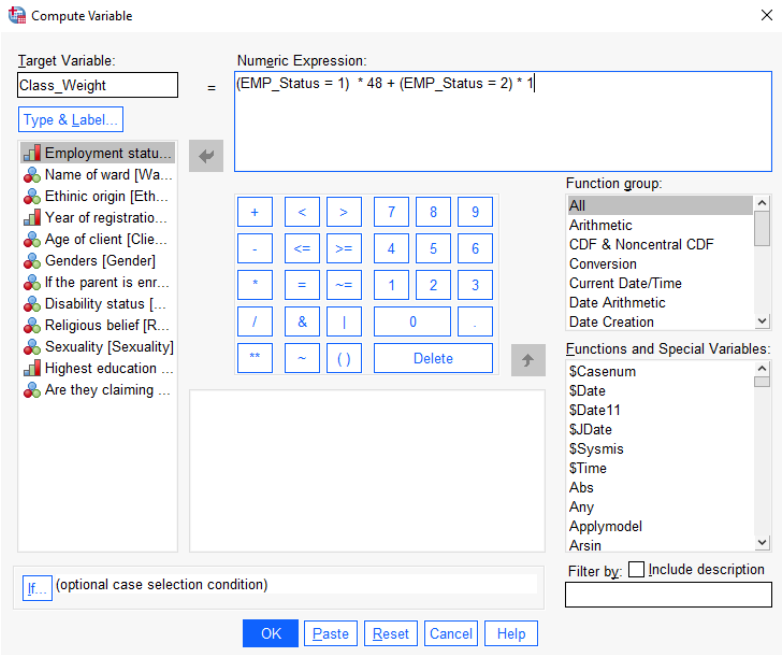
Variables in the Equation

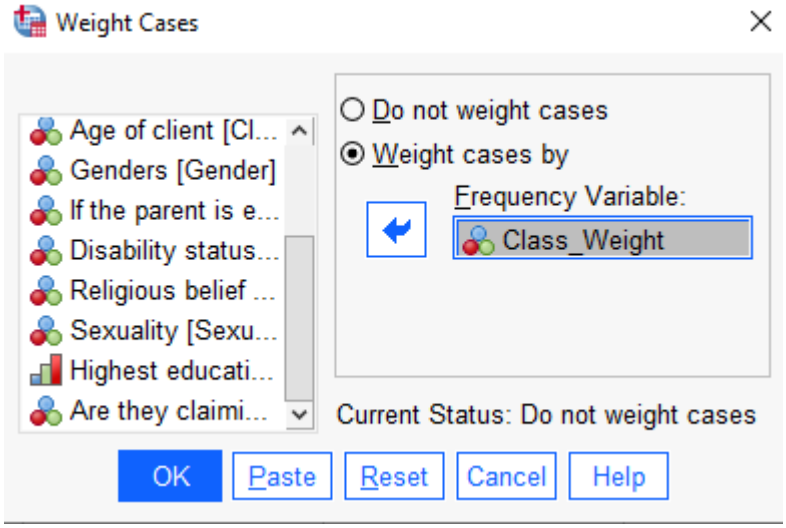
		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1[a]	Year of registration	.783	.049	259.354	1	<.001	2.187
	If the parent is enrolled	.251	.056	20.327	1	<.001	1.285
	Disability status	.282	.042	45.918	1	<.001	1.326
	Highest education level	.106	.019	31.868	1	<.001	1.112
	Constant	-1580.356	98.072	259.666	1	<.001	.000

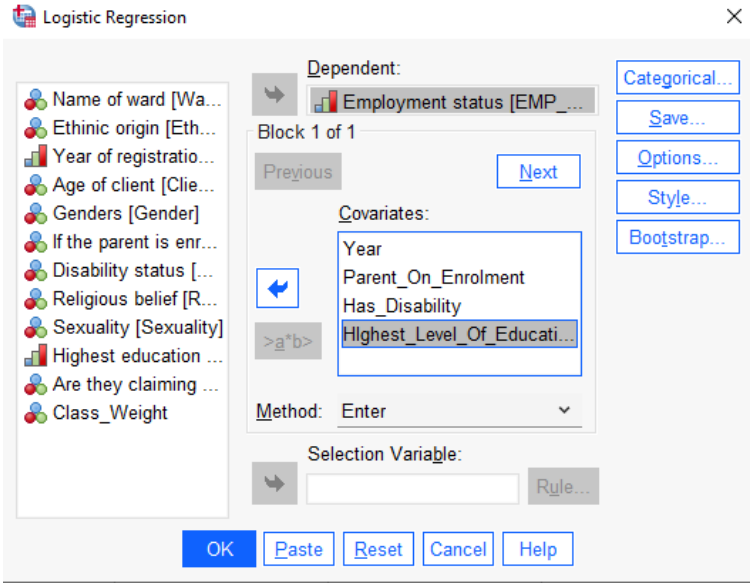
a. Variable(s) entered on step 1: Year of registration, If the parent is enrolled, Disability status, Highest education level.

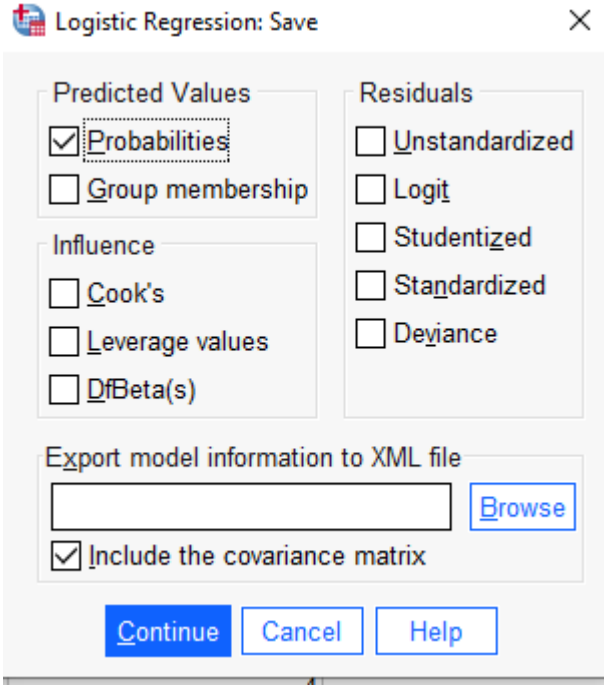
Model evaluation using the ROC-AUC curve

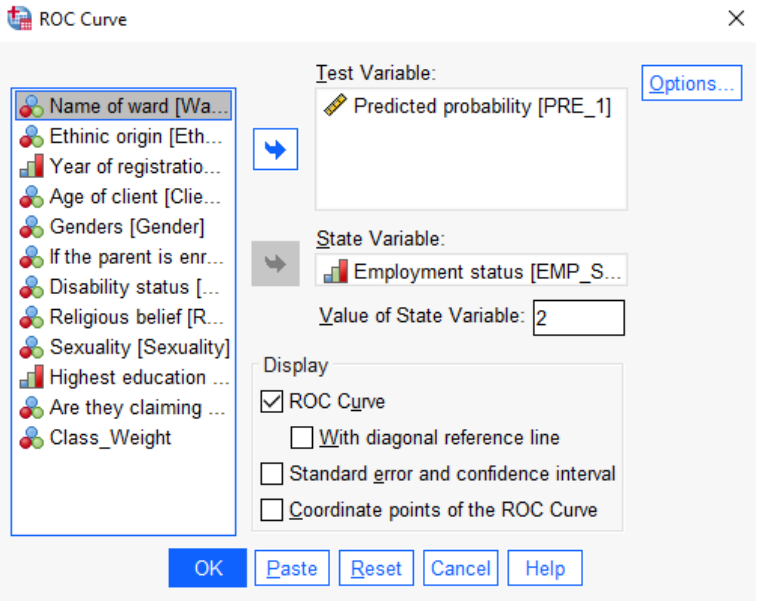
I have just realised the problem with the massive difference in unemployed and employed. To try and solve this I will add more weight to the "Employed" value.

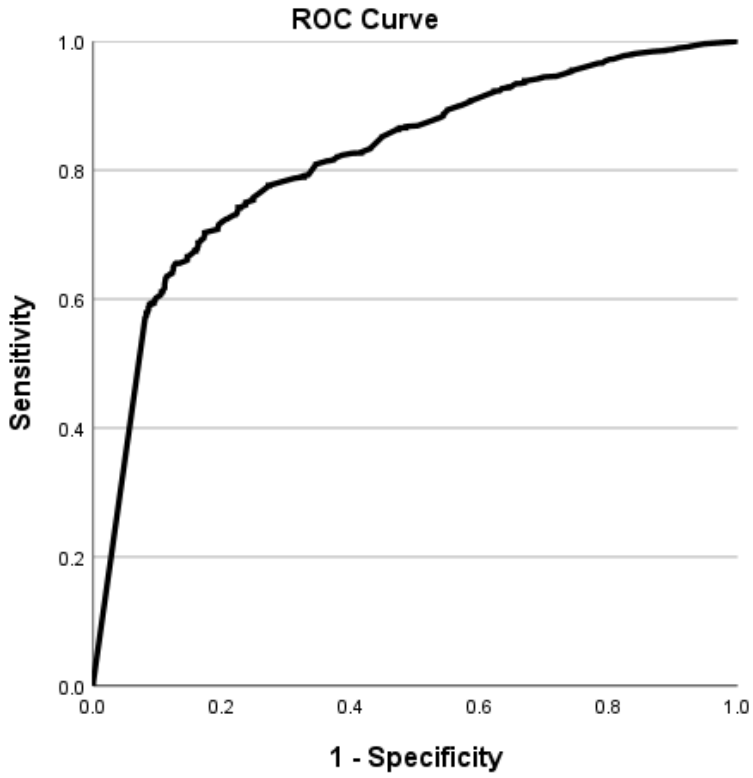



Weight Cases
Age of client [Cl...
Genders [Gender]
If the parent is e...
Disability status...
Religious belief ...
Sexuality [Sexu...
Highest educati...
Are they claimi...
Do not weight cases
Weight cases by
Frequency Variable:
Class_Weight
Current Status: Do not weight cases
OK
Paste
Reset
Cancel
Help


Logistic Regression
Name of ward [Wa...
Ethinic origin [Eth...
Year of registratio...
Age of client [Clie...
Genders [Gender]
If the parent is enr...
Disability status [...
Religious belief [R...
Sexuality [Sexuality]
Highest education ...
Are they claiming ...
Class_Weight
Dependent:
Employment status [EMP_...
Block 1 of 1
Previous
Next
Covariates:
Year
Parent_On_Enrolment
Has_Disability
Highest_Level_Of_Educati...
>a*b>
Method: Enter
Selection Variable:
Rule...
Categorical...
Save...
Options...
Style...
Bootstrap...
OK
Paste
Reset
Cancel
Help


Logistic Regression: Save
Predicted Values
Probabilities
Group membership
Influence
Cook's
Leverage values
DfBeta(s)
Residuals
Unstandardized
Logit
Studentized
Standardized
Deviance
Export model information to XML file
Browse
Include the covariance matrix
Continue
Cancel
Help


ROC Curve
Name of ward [Wa...
Ethinic origin [Eth...
Year of registratio...
Age of client [Clie...
Genders [Gender]
If the parent is enr...
Disability status [...
Religious belief [R...
Sexuality [Sexuality]
Highest education ...
Are they claiming ...
Class_Weight
Test Variable:
Predicted probability [PRE_1]
Options...
State Variable:
Employment status [EMP_S...
Value of State Variable: 2
Display
ROC Curve
With diagonal reference line
Standard error and confidence interval
Coordinate points of the ROC Curve
OK
Paste
Reset
Cancel
Help



Diagonal segments are produced by ties.

Area Under the Curve

Test Result Variable(s): Predicted probability

Area
.818

Discussion and interpretation of the results

Starting with the logistic regression analysis I have left out "Client_Current_Age" as I have found that it is highly correlated with other variables, and this will skew the results. Looking at the p values (Sig) we can see that the statistically significant ones are as follows, "Year of registration" (<.001), "Disability status" (<.001), "If the parent is enrolled" (<.001) and "Highest education level" (<.001) all the other values are non-significant and can be removed. The Exp(B) column shows us a positive relationship with the target variable with "Year of registration" being the strongest, later years may indicate better employment chances.

Now for the ROC-AUC curve, the area under the curve is 0.818, which indicates that the model is effective in distinguishing between "Employed" and "Unemployed" with a 81.8% accuracy this is also backed up by looking at the graph, we can see that it has a steeper curve near the top left corner which indicates better performance.

Conclusion

In conclusion, this coursework aimed to predict employment struggles within the Islington area using different analysis techniques such as logistic regression using various metrics such as the ROC curve. The initial stages of the coursework such as the data preparation were straightforward and manageable. However, as I progressed through the coursework I encountered challenges. The main difficulty arose from dealing with a highly imbalanced target

variable, despite these challenges I was able to use the knowledge I used from the module to navigate through the process, applying techniques such as class weighting to address the imbalance in the data. The ROC curve showed that the model was able to distinguish between “Employed” and “Unemployed” in the target variable with a good degree of accuracy, however as this is my first time undertaking a data set like this, I am sure there are some things that I have done wrong or inefficiently that may have changed the results. This coursework provided a valuable insight into the sort of data analysis that is expected in day-to-day life, and I feel like I have a better understanding of SPPS and data analysis as a whole.